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Graduation Project Documentation

WI-FI Controlled drone

**A Project Submitted in partial fulfilment of the
requirements for the Degree of Bachelor of Science in
Computer Engineering**

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ABSTRACT

A drone is a kind of aircraft and known as an unmanned aerial vehicle (UAV). The “unmanned” name given to drones because it does not need a pilot on board to fly it. Some drones are either guided by a remote control, by computers, or by people thousands of miles away. Drones are available in different sizes and shapes. Since early 2013, people have started flying drones for fun and that caused the governments to create rules for flying them. During the Civil War (1861–1865), armies tried using balloons to drop explosives into enemy camps. The British Royal Navy created a drone-like vehicle in the 1930s. Also, pilots used it to practice their shooting skills. However, drones are safer to fly over dangerous areas to collect information or in finding hidden places for enemies.

In this project, a WIFI controlled drone is going to be designed and implemented using a flight controller, ESP32, and a combined software of Betaflight, Blynk platform, and Arduino C++ compiler. The essential components such as flight controller, controlling system, power delivery, and drone components are covered. The flight controller is going to be configured using Betaflight software and Blynk platform which is going to be used to set up the buttons and slide buttons for the speed of the brushed rotors to control the drone direction or movement based on the required speed that initiate torques around the drone and consequently control movement. Also, building Arduino C software with required speeds of each motor through WIFI. Home router or WIFI of cellphone can be used to connect the drone with the Blynk platform via internet. It is worth mentioning that the processing speed depends on the series of controller used. (i.e. F3, F4, or F7). Also, ESP32 is connected with the flight controller receiver to receive commands and execute them via flight controller to control the rotors speed and consequently the drone movement. Cellphone or IPAD is going to be used to send the commands via Blynk platform.

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GLOSSARY AND LIST OF ABBREVIATIONS

Design Constraints:

- 1.UAV: Unmanned Aerial Vehicles
- 2.VTOL: Vertical take-off and landing
- 3.FBW: Fly by wire
- 4.YAW: Pitching motion
- 5.TMC: Total Moment Calculation
- 6.BLDC: Brushless DC Motors
- 7.RPM: Revolutions Per Minute
- 8.ESC: Electronic Speed Controller
- 9.FET: Field-Effect Transistor
- 10.MOSFET: Metal-Oxide-Semiconductor Field-Effect Transistor

- 11.JFET: Junction Field-Effect Transistor
- 12.BEC: Battery Elimination Circuit
- 13.UBEC: Universal Battery Elimination Circuit
- 14.NiCad: Nickel-Cadmium
- 15.NimH: Nickel-Metal Hydride
- 16.LiPo: Lithium Polymer
- 17.LiFePo: Lithium Iron Phosphate
- 18.JST: Japan Solderless Terminal
- 19.GPS: Global Positioning System
- 20.WI-FI: Wireless Fidelity
- 21.UART: Universal asynchronous receiver-transmitter
- 22.Rx/Tx: Receiver and transmitter
- 23.PWM: Pulse width modulation
- 24.PPM: Pulse Position Modulation
- 25.sBUS: Serial Bus
- 26.iBUS: Intelligent Input Bus
- 27.Baud rate: times signal changes in a second
- 28.Bit rate: bit sent per second
- 29. KV: constant voltage

Chapter 1

Introduction

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1.1 INTRODUCTION

A drone, also known as an unmanned aerial vehicle (UAV), is an unmanned aircraft that can be controlled remotely using sensors and GPS. Originally associated with the military and intelligence gathering, drones have evolved to serve various purposes, including surveillance, firefighting, traffic and weather monitoring, search and rescue, and personal tasks. They are now integral in industries like filmmaking and journalism for capturing footage in remote locations. A UAV system comprises the drone and its control system.

1.2 History of drones

Drone history is believed to start about 172 years ago, since then it evolves in many sectors, largely military, then the other sectors adopted it, military was largely response for the advancement of the drones as a technology from torpedoes to UAVs. The siege of Vienna in 1849 marked the first time balloons were used in an organized warfare when more than hundred balloons were launched from an Austrian ship to force the Venetians to surrender. The U.S UAV program began in 1959 when U.S air force planned to use unmanned aircraft, since 1964 the technology has advanced further. First unmanned vehicle was developed by US/UK in WW1 that was small radio-controlled aircraft and it was tested in 1917, while American torpedo made its first test in 1918. The term "drone" was believed to be used in 1935 after the UK multiple drone's aircraft to be used as targets for training. It was also used in the US for the same purpose. However, drones in our current time are used in many professions such as education, agriculture, filming and of course military. In the 2022 war between Ukraine and Russia, Drones were used in attack, surveillance, clearing mines, etc. Also, it is widely used in around the world by people usually for photography and usually requiring only a license. By providing cutting-edge functions, lowering the need for human interaction, and lowering operational costs. At the same time, drone technology has had a substantial impact on several industries. However, just as with any technology, using them comes with risks like safety concerns, potential misuse, regulatory complexity, and privacy issues. Balancing between these difficulties and the potential advantages of drone

technology continues to be a key topic of discussion. Despite this, the future of drones seems promising as ongoing developments increase their possible applications [1].



Figure.1.1: History of drones [2]

1.3 Drone Technology versatility

Drones, also known as Unmanned Aerial Vehicles (UAVs), have their origins in science fiction and represent unmanned aircraft. They operate at the intersection of aerospace, robotics, and mechatronics and have diverse applications, including filming, military surveillance, and recreation. Drones are remotely controlled flying robots that are integral to businesses and government agencies, enabling rapid delivery and access to remote locations. They can be piloted through remotes or mobile apps and excel in conditions deemed dry, dull, or dangerous for human pilots. This technology has gained global adoption due to its versatility and benefits.

1.4 How Do Drones Work?

Drones are sophisticated aircraft systems with two main functions: flight and navigation. They are powered by batteries or fuel and consist of rotors, propellers, and lightweight frames for maneuverability. A remote controller communicates with the drone using radio waves, enabling launch, navigation, and landing. Key components of drones include flight controllers, accelerometers, barometers, magnetometers, gyroscopes, and data collectors for stability. Additional features such as GPS, obstacle

detection, cameras, and software enhance performance. Drones are essentially flying robots, controlled remotely and guided by various sensors, making them versatile and adaptable for various applications.

1.5 Parts of the Drones

A drone comprises ten essential parts. These are as follows:

1. Quad-copter frame: The structural base that houses all other drone components.
2. Motor: Generates rotational force to drive propellers and propel the drone.
3. Electronic Speed Controller (ESC): Controls motor speed and functions as a dynamic brake.
4. Flight Control: Records the takeoff location and guides the drone, including the "back home" feature.
5. Propeller: Creates a difference in air pressure to enable drone flight.
6. Radio transmitter: Channels communication with the drone.
7. Battery, electronics, and power distribution cables: The power source for the drone, distributing power to onboard electronics.
8. Camera: Captures video footage for recording, storage, and transmission.
9. Landing gear: Safely facilitates drone landings, with experienced users able to control motor speed for emergencies.
10. First-person video (FPV): Provides an immersive 3D viewing experience through an interface, enhancing the user's sense of flight.

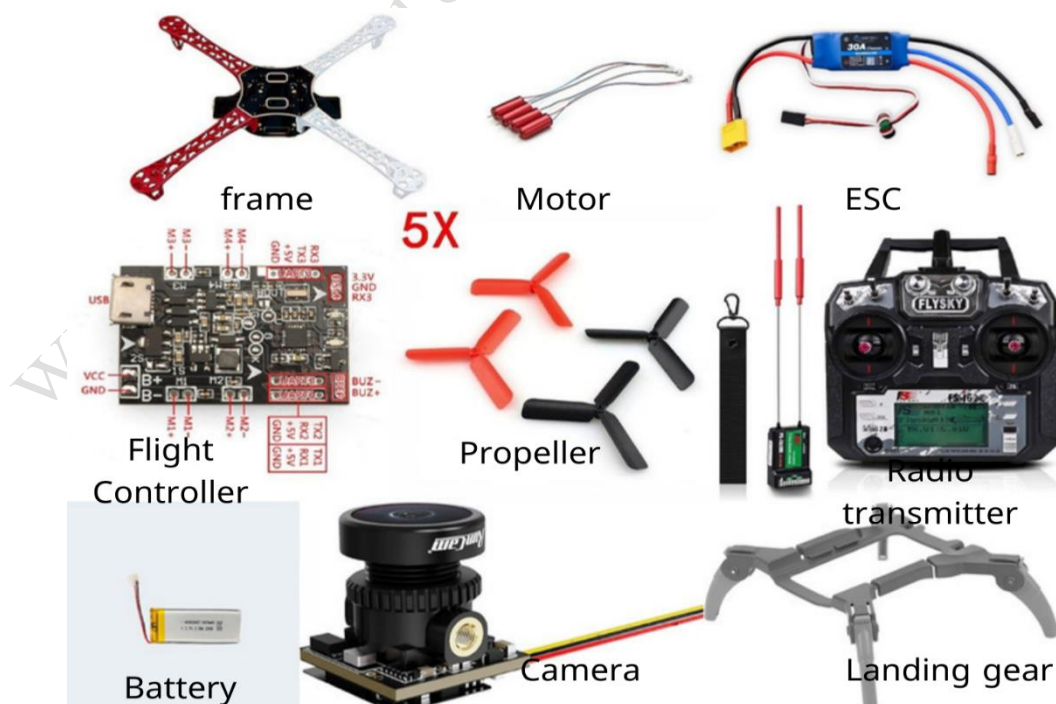


Figure.1.2: Parts of the drones

1.6 Types of drones:

There are more than a dozen types of drones. These can be categorized as follows:

1.6.1 single-rotor drone

These drones are robust and resemble real helicopters in design, featuring a single rotor, also known as a tail unit.



Figure.1.3: Single-rotor drones [4]

1.6.2 Multi-rotors drones

Multi-rotors Drones come equipped with four rotors, which makes them stable in the air but limits their flight duration to approximately thirty minutes. They are particularly useful for tasks like aerial photography and surveillance. However, the number of rotors affects the drone's flight time, with more rotors having a shorter airborne capability.



Figure.1.4: Multi-rotor drones [5]

1.6.3 Fixed wing drone

Has one grid wing that is designed to work like an airplane fixed wing drone can stay in the air for up to 16 hours, fixed-wing drones can be gas engine powered.



Figure.1.5: Fixed wing drone [6]

1.6.4 Fixed-Wing Hybrid Drones

Hybrid VTOL drone types merge the benefits of fixed-wing and rotor-based designs. This drone type has rotors attached to the fixed wings, allowing the drone to rotate and land vertically.



Figure.1.6: Fixed wing Hybrid drone [7]

1.6.5 Small drones

Small drones are generally less expensive; they cannot perform commercial functions that other drone models carry out. These drones are too light.



Figure.1.7: Small drones [8]

1.6.6 Micro drones

Micro drones called the Black Hornet. These types of drones can fly up to 25 minutes in single charge and have a range of up to one mile.

1.6.7 Tactical drones

Tactical drones are larger than small drones and are used for military purposes. They can be used for surveillance, bombing, or other combat operations.

1.6.8 Reconnaissance drones

Reconnaissance drones are used for military purposes. They are typically used to gather intelligence on enemy positions or movements.

1.6.9 Large combat drones

are the largest and most powerful type of drone. They are used for long-range (over 1,000 miles) strikes or bombing missions.

1.6.10 Non-combat large drones

Non-combat large drones are used for surveillance or other non-combat purposes where they are not armed.

1.6.11 Target and decoy drones

Target and decoy drones are used to simulate enemy targets. They can be used for training or for testing weapons systems.

1.6.12 GPS drones

GPS drones use GPS to navigate. They are often used for mapping or surveying.

1.7 Applications of Drone Technology

Drones are now used in many different applications ranging from military to civil, drones now make our life easier and safer, now drones are used by most professions,

and they constantly improved either in the technology or in the profession they are used in. Drones enhance any industrial facility they are used in, offering cheap and effective aerial perspectives. Drone's usage potential in industry are endless. Industrial professionals are increasingly using drones to improve industrial processes. Drones are uniquely suited to collect aerial data to use in large industrial facilities, ensuring rapid data collection. Automated drones can be used in industrial facilities for surveillance and security, weather monitoring, equipment monitoring etc. Drones can also be used for mapping and border monitoring [9].

1.7.1 Aerial Photography

Drones now can be equipped with advance cameras to be used for media coverage, filming, aerial photography, site surveying, and 3D imaging. While helicopters were once limited to large news companies, drones now enable small media and local journalists to capture impressive aerial footage. Drones' small size allows them to access tight spaces and even record face-to-face interviews.



Figure.1.8: Aerial Photography

1.7.2 Agriculture

Drones have revolutionized the field of agriculture by providing numerous benefits to farmers. They are extensively used for various purposes, making agriculture one of the key areas for commercial drone applications. Drones enable farmers to detect failing crops, monitor irrigation systems, and analyze performance, resulting in cost savings and improved crop health. Additionally, drones provide timely updates and assist in precise spraying of fertilizers, pesticides, and water, ensuring optimal crop care.



Figure.1.9: Agricultural drones

1.7.3 Search and rescue

Drones can be very helpful in situations where you need night vision to locate missing or lost people. Also, it can go to places where it is hard and unsafe for others to rescue them. These can be shown in Figures 10 and 11 respectively.



Figure.1.10: Night vision taken by drone.



Figure.1.11: Search and rescue drone

1.7.4 Science and Research

Scientific researchers use drones to gather data in dangerous or difficult-to-reach places like deep-sea or volcanic areas. They are also used for mapping, environmental surveillance, and animal monitoring.



Figure.1.12: Science and Research drone

1.7.5 Building and construction inspections

Drones can collect a large range of data and can capture high-resolution, thermal images and 3D imaging. Also covers wide Areas where it takes days or weeks to inspect completely, and it can be done with drones within hours only. This will help the business become more efficient by saving time and money.



Figure.1.13: Construction inspections drone [10]

1.7.5 Military Drones

The military has long employed drones for surveillance, combat operations, and spying. Modern, advanced drones are equipped with weapons and thermal image. They assist in lowering casualties while successfully completing tasks.



Figure.1.14: Military Drone [11]

1.7.6 Weather forecast

Drones can collect detailed atmospheric data and reach heights and locations that are hard for other weather technologies to reach. This can increase the accuracy of weather forecasting.



Figure.1.15: Weather drones [12]

1.7.8 Shipping and delivery

Drone delivery offers fast, efficient shipping to remote areas, bypassing traffic. It saves costs, reduces emissions, and improves last-mile delivery.



Figure.1.16: Shipping and delivery drones [13]

1.7.9 Entertainment drones

Drones are used in the entertainment industry in drone racing, concerts, sporting events, and filmmaking. They can be programmed to fly in synchronized formations, creating mesmerizing patterns and shapes in the sky.



Figure.1.17: Drone in entertainment industry [14]

Chapter 2

Drones Aerodynamics

2.1 Introduction

Rotating wings were early designed but were initially surpassed by autogyros and helicopters due to inherent stability issues. However, multirotor drones have become popular due to their distinct advantages. In helicopters, the tail rotor compensates for reaction torque but doesn't contribute to thrust. In contrast, multi-copters with four rotors can efficiently harness power for vertical thrust, making them more efficient. The stability challenges of multi-rotor designs were resolved with the introduction of solid-state gyroscopes, accelerometers, and magnetometers. These electronic components enable precise measurement of angular positions, acceleration, and magnetic fields. Flight systems for quadcopters, like Arduino, combine these technologies. Sensor data is processed by a microprocessor using cascading regulation loops, allowing pilot inputs to work in harmony with the flight control system, akin to "fly by wire" (fbw) systems in modern fighter aircraft.



Figure.2.1 Mid-20th century quadcopter [15]

Quadcopters are aircraft with multiple rotating wings, governed by FBW systems, offering both pilot control and advanced self-stabilization and flight path correction capabilities. Figure 2.1 shows mid 20th century quadcopters.

2.2 Aerodynamic Principles of Functioning of a quadcopter

The principles governing the movement of a quadcopter in three spatial axes are explained, particularly focusing on a quadcopter with an X-shaped frame. Pitch (for

movements on the lateral axis), roll (for rotations around the longitudinal axis), and yaw (for movements around the vertical axis) are the definitions of these axes of rotation. Figure 2.2 shows a conventional aircraft with its three axes of displacement.

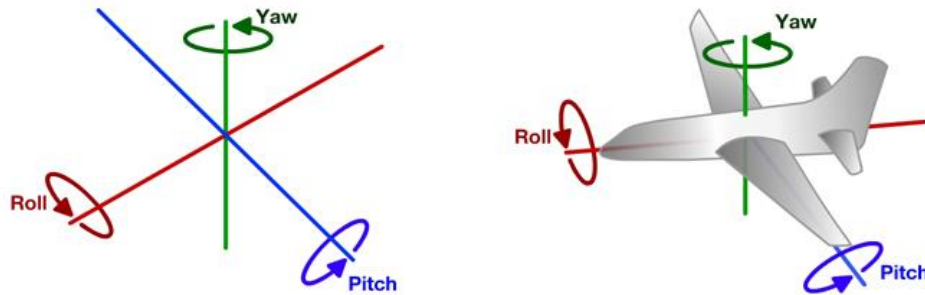


Figure.2.2 Conventional aircraft with its three axes of displacement [16].

In the case of a multirotor aircraft, the names of these movements and the axes remain the same, with a forward pitch corresponding to a forward displacement and vice versa. Similarly, right and left roll movements result in corresponding lateral movements.

To understand the operation of a multirotor, it's crucial to grasp the concept of reaction pairs as per Newton's third law. The explanation centers on the torque produced by a motor turning a propeller, creating an equal and opposite reaction pair on the motor's frame. These reaction pairs are critical for maneuvering the aircraft.

The movements of a multirotor are achieved by compensating or decompensating the reaction pairs generated by the different rotors, creating a resultant force in the desired direction. The analysis begins with the simplest scenario: level stationary flight.

2.3 Level flight

A quadcopter with an X-shaped configuration employs two motors rotating clockwise and two motors rotating counterclockwise. The propellers have antisymmetric blade designs, but all of them generate lift despite their opposite directions of rotation. To maintain stable stationary flight, all four motors rotate at the same speed, ensuring that the reaction pairs on the four motors cancel each other out. This balance prevents the quadcopter from rotating around its vertical axis. However, practical factors like aerodynamic effects, residual asymmetries, and atmospheric conditions necessitate real-time control and stabilization system adjustments to maintain stable stationary flight.

2.4 Forward and backward pitch (forward and reverse)

Stationary Flight: In stationary hover, all rotors provide equal thrust, keeping the quadcopter level and stable at a certain height. Real-time control and stabilization are necessary due to external factors like aerodynamic effects.

Forward Pitching: To initiate a forward pitch, thrust is increased equally in the tail rotors or decreased equally in the nose rotors. This tilts the quadcopter forward and initiates forward motion. The control system maintains height during this process.

Backward Pitching: Backward pitching is the reverse of forward pitching, causing the quadcopter to move backward.

Right and Left Warping (Lateral Displacement): To initiate lateral movement, the pair of rotors on one side of the quadcopter increase their speed while the other pair decreases it. This results in a rotation around the longitudinal axis, causing lateral displacement. Again, the control system ensures height is maintained during this movement.

In all these movements, it is essential to maintain equilibrium in terms of thrust to counteract gravity (TQM) and adjust the quadcopter's angular acceleration. The quadcopter's behavior is governed by Newton's third law and torque principles, with differential motor thrusts creating the desired movements. These movements are controlled and stabilized in real time to maintain stability and height during flight.

2.5 Right and left warping (lateral displacement)

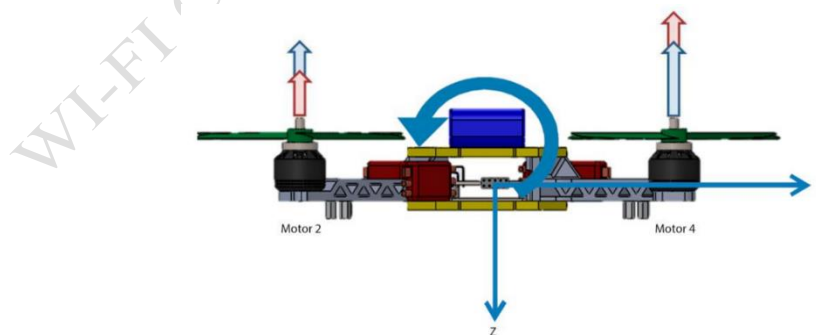


Figure.2.3 Right and left warping [17]

Warping, the mechanism for lateral displacement in quadcopters, is akin to pitch movements but involves different pairs of motors. When the quadcopter needs to move to the right, the right-side motors increase their lift, and the left-side motors compensate by also increasing their lift, causing a rotation along the longitudinal axis to the right. The process is reversed for warping to the left.

During warping, the vertical thrust vector is divided into two components: one counters the weight of the aircraft, and the other induces lateral movement, resulting in the quadcopter's lateral displacement.

The equilibrium equation for the z-axis remains balanced during this state. When applying the Total Moment Calculation (TMC) forces T_1 and T_3 counterbalance, creating angular momentum. If there's a difference between T_2 and T_4 , a moment is generated along the positive y-axis, leading to angular acceleration, as defined by the equation:

$$\ddot{\phi} = \frac{d(T_4 - T_2)}{l_{yy}} \quad (1)$$

This also induces horizontal translation in the direction in which the drone has been inclined.

2.6 Stag flight in level stationary flight (turn on itself)

Quadcopters utilize various techniques to control yaw (rotation around the vertical axis):

Opposing Rotors: This common method involves pairs of rotors spinning in opposite directions to balance torque. It requires two types of propellers due to their aerodynamic profiles.

Tilting Rotors: An alternative approach is to tilt all rotors to counteract torque. All rotors spin in the same direction, but their thrust is angled to offset the torque. Yaw control is achieved by adjusting the speed of two of the four rotors.

Variable Rotor Orientation: In some cases, a rotor is allowed to rotate perpendicular to the arm using a servomotor, enabling yaw control in different directions. This method is more complex and is often used in tri-rotors.

Control Surfaces: Control surfaces at the rotor outlets can redirect airflow to provide counter-torque by changing the angle of these surfaces.

Generally, the first method (opposing rotors) is preferred for its mechanical and aerodynamic simplicity. Yaw control involves modifying the behavior of rotor pairs to create a resultant force for rotation. For example, counterclockwise rotation is achieved by reducing the speed and lift of specific rotors while increasing others. This differential rotor behavior enables yaw motion without losing altitude. This is shown in Figure 2.4. The maximum speed of a quadcopter depends on thrust and air resistance. Maximum linear acceleration can be calculated based on the maximum slope the quadcopter can handle while maintaining its altitude, using specific equations and parameters.

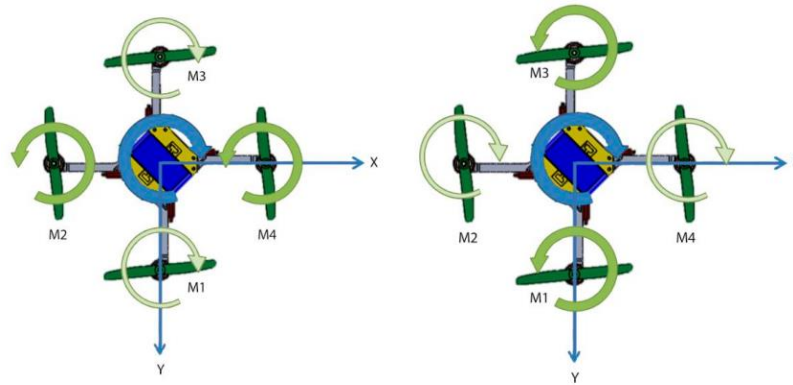


Figure.2.4 Yaw movement of quadcopters [17]

It's worth noting that quadcopters are generally less efficient in yaw control compared to helicopters due to their limited means of generating torque, resulting in slower yaw rotations.

2.7 Types of chassis or frames

The structures in the accompanying diagram correspond to the three kinds of quadcopter frames, designated as X, H, and +:

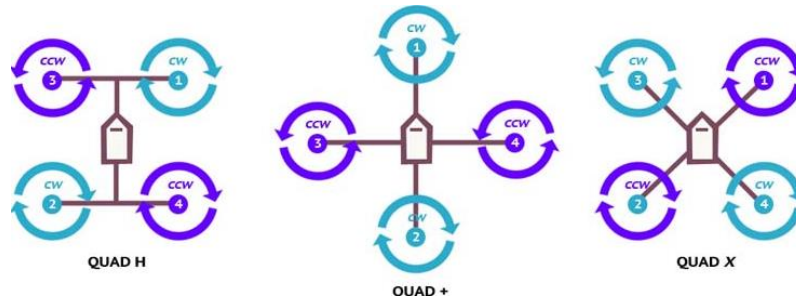


Figure.2.5 The three types of quadcopter frames [18]

the behavior and operation of a quadcopter are not significantly affected by the type of chassis used. The H and X configurations are functionally identical, with differences mainly in the ease of arranging and accommodating equipment and the structural strength. The H-structure is preferred for carrying larger and heavier equipment and is commonly used in commercial quadcopters.

Commercial carbon fiber frames often use the following equipment in X configurations:



Figure 2.6 Carbon fibre frames [19]

the + configuration for quadcopters is less commonly used. Structurally, it is similar to the X configuration, and you can switch between X and + with the same frame, but the control and maneuvering procedure is different. Figure 9 illustrates that pitching movements are achieved by varying the lift of the front and rear engines, and roll movements occur when the lateral motors' lift is adjusted. Unlike X or H configurations, where all four rotors contribute to thrust differences for movement, the + configuration relies on just two rotors for forward, backward, and lateral movements. This makes the + configuration less efficient for maneuvering.

2.8 Propellers

the most crucial part of a quadcopter is the propeller. It is a revolving wing with blades that are designed for aerodynamics. It is responsible for converting the power supplied by the engine into useful thrust for flight. The propeller cross-section shown in Figure (2.7) is shown at 75% of its diameter. The edge that leads is referred to as the top edge, and the trailing edge is the bottom edge. The line connecting the leading edge to the air (the relative wind) forms the angle of attack. Vertical thrust is produced when the propeller is moving horizontally and creating more pressure at the back than the front. This can be depicted by the torque imposed by the resultant forces on the propels as can be shown in figure 2.8.

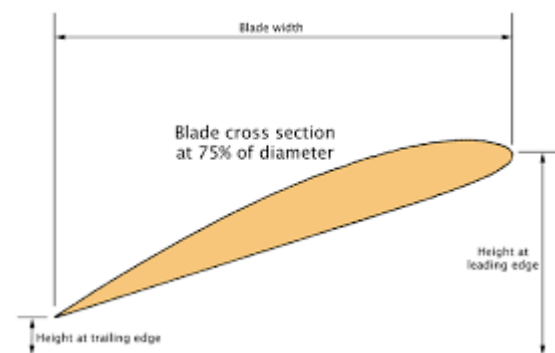


Figure.2.7 Profile of a propeller [17]

The propeller blades are twisted from the tips towards the center due to the inverse correlation between an airfoil's lift and its relative airfoil velocity. The tips require a lower angle of attack to generate the same amount of lift as the center, as they rotate more quickly. For tiny, swiftly rotating propellers, this twist is particularly crucial. The tips of the propeller can go close to the speed of sound at these speeds. When this occurs, the air becomes compressible, which changes the equations used to determine lift and drag. The propeller's tips are made with negative angles of attack to make up for this. A pair of puller-type propellers and another pusher-type propeller are needed in a quadrotor type helicopter. The names describe the rotational direction that the propeller must take to produce a pushing force that will assist the vehicle's flight. depicts several factors that can affect any propeller, and which must be understood. the thrust it produces, the torque that rotates it, and the sum of the parts that make up A and B.

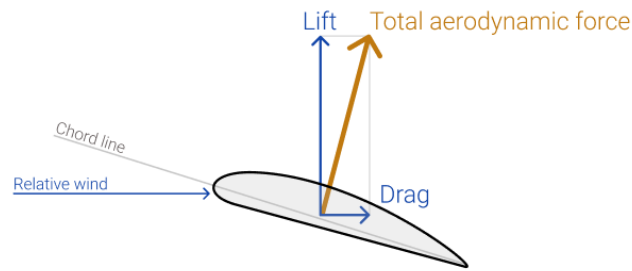


Figure.2.8 Forces of flight [20]

2.9 selecting propeller

When selecting a propeller for a quadcopter design, the following elements should be taken into consideration:

- a) rigidity and manufacturing quality: The best material for the propellers would be carbon fiber or wood, and they should be as stiff as they can be. Flexible propellers will warp when loaded, reducing flying performance and stability.
- b) Diameter: The better propellers have a bigger diameter. This is because a bigger propeller will provide more thrust and operate more effectively. Due to the limited amount of available space, the propeller diameter should be selected to maximize the area that it can cover without interfering with the rotors.
- c) Pitch: Pitch of propeller impacts thrust, with higher pitches requiring more power, while spinning propellers consume more power. Weight and desired performance should be considered.

Tandem vortex interference: When two propellers are too near to one another, a phenomenon known as tandem vortex interference takes place. The propellers' effectiveness is decreased as a result of the interaction between their vortices, which also makes the quadcopter more challenging to control.

To prevent tandem vortex interference, the distance between the motor axes on a quadcopter should be at least 1.25 times the diameter of the propellers.

2.10 BRUSHLESS DC MOTORS (BLDC)

The BLDCM, despite its complexity, offers several advantages over its counterpart, including higher torque/weight ratio, less operational noise, longer lifetime, less electromagnetic interference generation, and more power per volume. However, its inherent heat generation is limited by conduction. Despite these differences, the BLDCM's dynamic model can be roughly approximated by the BDCM's. The rotor and the stator are the two components that make up a BLDC motor. Of course, the first one moves, whilst the second one is still. The rotor of an electric motor is typically a hollow cylinder that rotates inside a larger hollow cylinder that serves as the stator. However, there are two alternative topologies for this kind of machine: "Inrunner and Outrunner" engines. These designations, which cannot be directly translated into Spanish, merely state whether the rotor is within the stator (inrunner) or whether it is outside the stator and the stator is fastened inside (outrunner).

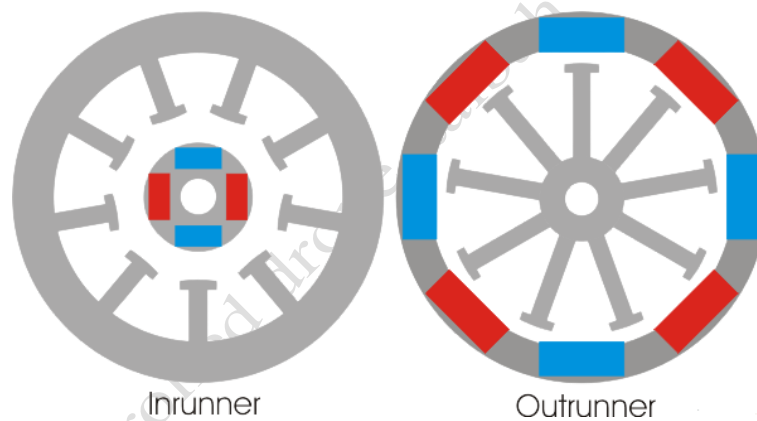


Figure.2.9 Outrunner and Inrunner Brushless Motors [21]

Differences Between Outrunner and Inrunner Brushless Motors:

- Physical size

In comparison to inrunner motors of comparable weight, brushless outrunner motors often have a bigger diameter and shorter length. In contrast, inrunner motors have a smaller diameter and usually have a longer length.

- RPM per volt

The inrunner motor often has a high speed per volt. The outrunner motor, on the other hand, has a low speed per volt while being the same size.

- Torque

Brushless outrunner motors typically have more torque than inrunner rotors due to their lower speed, which is inversely proportional to rotational speed, meaning that as speed increases, torque decreases.

- Efficiency:

Brushless inrunner motors typically have lower inertia than brushless outrunner motors, whereas outrunner motors have more inertia and are less efficient.

2.11 Principle of operation: electronic switching

motors are based on electronic switching, which involves the injection and shutdown of current through the stator windings. When no current is applied, the rotor remains stationary. However, when current is introduced in a specific phase, magnetic poles are created, resulting in attraction and repulsion forces that cause the rotor to rotate. By periodically injecting current at the proper angular position, the motor achieves continuous rotation. This process is controlled by electronic switches within the variable speed drive, and the motor speed is adjusted by changing the frequency of current activation and deactivation. In practical applications, BLDC motors often incorporate external systems such as encoders to provide the rotor's angular position to the speed controller. However, sensor less control algorithms can also be used when external position feedback is not available. Sensor-less control relies on the principles of electromagnetic induction, where the movement of a magnetic field induces electromotive forces in the motor's coils. By measuring these induced forces, the speed controller can estimate the rotor position and synchronize the electronic switching accordingly. BLDC motors are characterized by parameters such as the kV constant, which indicates the RPMs expected for a given supply voltage. The torque and speed of the motor are related through equations that involve the supply voltage, speed, torque, and current consumed. The no-load current represents the motor's consumption to overcome friction losses and magnetization. Other parameters include the resistance of

the windings, maximum permissible current, electrical power absorbed, and the number of poles.

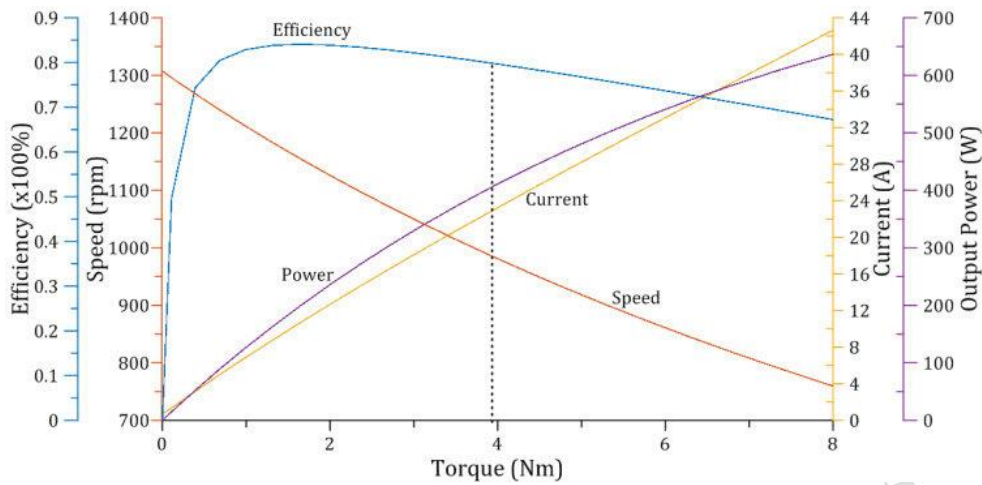


Figure.2.10 Characteristic curves of the designed BLDC motor [22].

The performance of a BLDC motor is represented by characteristic curves, where the torque, speed, and efficiency are plotted against the operating point. The operating point should be chosen close to the point of maximum performance to ensure efficient power utilization and prevent overheating as can be shown in figure 2.10. It is important to select propellers that bring the motor close to this point and discard combinations that yield low efficiencies. Operating temperature is also a crucial consideration for BLDC motors, as the flux density of permanent magnets decreases with increasing temperature. To reduce engine heating, the design should take this into consideration. Around 40 °C would be an ideal working temperature, and 60 °C should never be exceeded.

2.12 Electronic Speed Controllers

An electronic speed controller (ESC) interfaces with brushless electric motors, converting a low-power microprocessor signal into a high-power three-phase alternating signal. The signal, typically a pulse width modulation (PWM) signal, determines the frequency and duty cycle, usually ranging from 1 to 2ms and 50hz to 490hz, respectively. The ESC's structure and operation can be summarized as follows Starting from the right side of the diagram in figure 2.11:

1. Output filter capacitor: It filters voltage wave oscillations applied to the motor.
2. Set of 6 switches labeled as FET: These low-resistance MOSFET-JFET transistors control the motor's phases by applying positive, negative, or zero voltage combinations.
3. FET Driver Circuitry: It adapts the microcontroller's logic outputs to the transistor gates, enabling their opening and closing.
4. Microcontroller: It governs the transistors using input signals such as throttle and measurements of induced electromotive forces in each phase.
5. BEC = Battery Elimination Circuit: It stabilizes battery voltage to power the microcontroller and other components requiring logical levels, typically 5V.

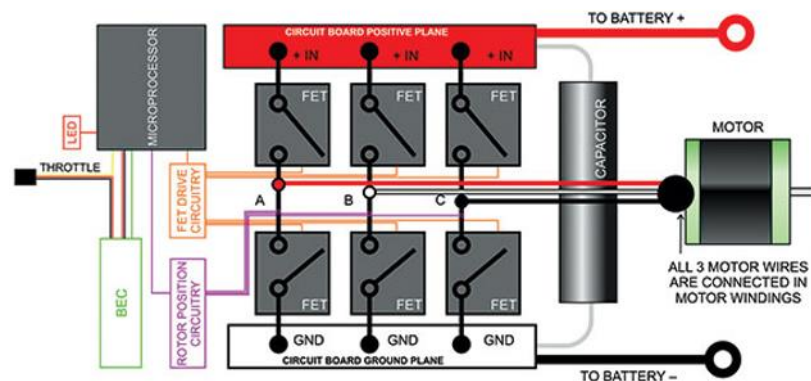


Figure.2.11 Simplified electronic speed control [23]

The ESC's operation involves the following steps:

1. The ESC is connected to the receiver equipment, and a calibration process is performed to set the gas channel input signal range and other operating parameters.
2. When the gas control is raised, the microcontroller receives the signal and adjusts the transistor switching frequency to achieve the desired speed.

When choosing an ESC, consider variables such as maximum current handling, battery voltage compatibility, presence of BEC, programming method, optocoupler for galvanic separation, and weight. The market offers a wide range of ESCs with diverse characteristics.

2.13 Battery Elimination Circuits

The BEC system, discussed earlier, provided power to the electronic components of electric models. However, linear regulators used in BECs faced limitations when handling high cell count batteries. To address this, UBECs were introduced as DC-DC reducing converters, offering higher current capacity and efficiency compared to linear regulators. In a 4S Lithium-Polymer battery setup, where each cell starts at 4.2V, a linear regulator in a BEC would need to dissipate significant power to reduce the voltage from around 17V to 5V for logic power supply. This inefficiency and heat generation become problematic when extracting considerable current. UBECs, on the other hand, are switched sources that can handle higher currents and efficiently convert battery voltage to power the system's electronic circuitry. They can be integrated within the variable speed drive or purchased separately as a standalone DC converter. Practically, UBECs have a straightforward application: the battery serves as the input, while the power connector delivers the adapted voltage to all electronic components. An example of a lightweight UBEC weighing about 5.6g, with a 5V output and 3A capacity, is shown in the figure 2.12.



Figure.2.12 UBEC

2.14 Battery selection

In the early days of electric modeling, NiCad batteries were commonly used but are now banned for domestic use due to heavy metal content. They had high weight, low

energy density, and limited discharge capacity. NimH batteries with higher capacities were introduced, but their charging and discharging processes generated excessive heat, and they still suffered from weight issues. Later, Lithium Polymer (LiPo) batteries, such as the Kokam brand, emerged in the late 90s. Initially, they had lower energy and discharge capacities compared to current LiPo batteries. However, since then, LiPo batteries and variants like LiFePo have seen significant development. Because of their lower price and better performance, they have become the preferred option. LiPo batteries differ from Lithium-Ion Batteries since they use a solid polymer electrolyte instead of a liquid one. They work by moving lithium ions back and forth between the anode and cathode metal plates. While LiPo batteries offer high energy density and discharge capacity, they do have certain drawbacks. Overcharging can cause them to expand, burst, or even catch fire. Discharging below a specific threshold can lead to chemical reactions, anode and cathode oxidation, and the risk of expansion and fire. Early use of LiPo batteries saw cases of fire due to charging failures or using damaged batteries. LiPo battery packs used in modeling and industrial applications consist of multiple cells connected in series or parallel series combinations. Each cell has a nominal voltage of 3.7V, which can reach 4.2V at full load. The discharge depth, the minimum voltage during use, should not fall below 2.8V. When selecting LiPo batteries, consider the following parameters that define commercial packs.

2.15 Parameters of Lipo Batteries

1. Number of cells: LiPo batteries are designated by the number of cells they contain, denoted as nSmP. For example, a 3S2P battery consists of two branches of three 3.7V cells connected in series and then connected in parallel. Common configurations include 1S, 2S, 3S, 4S, etc., where the overall voltage is obtained by multiplying the number of cells by 3.7V, we usually use 1S, 2S in small drones.
2. Capacity: The battery pack's capacity is indicated by the letter C and measured by (mAh). It represents the amount of energy the battery can deliver. For instance, a battery with a capacity of $C = 4000\text{mAh}$ can theoretically provide a

current of 4A for one hour, reducing its voltage slightly (down to the result of multiplying the number of cells by 2.8V, the minimum permissible voltage).

3. Discharge capacity: The discharge capacity of a LiPo battery is expressed in multiples of C and indicates the maximum continuous and short-term current that can be drawn from the battery. For example, a battery labeled as 40C-60C can be continuously discharged at 40 times its capacity ($40 * 4000\text{mA} = 160000\text{mA} = 160\text{A}$) and briefly discharged at 60 times its capacity ($60 * 4000\text{mA} = 240000\text{mA} = 240\text{A}$). However, the battery's actual capacity under these conditions is reduced due to decreased efficiency at higher discharge rates. These parameters can be depicted from figure 2.13.

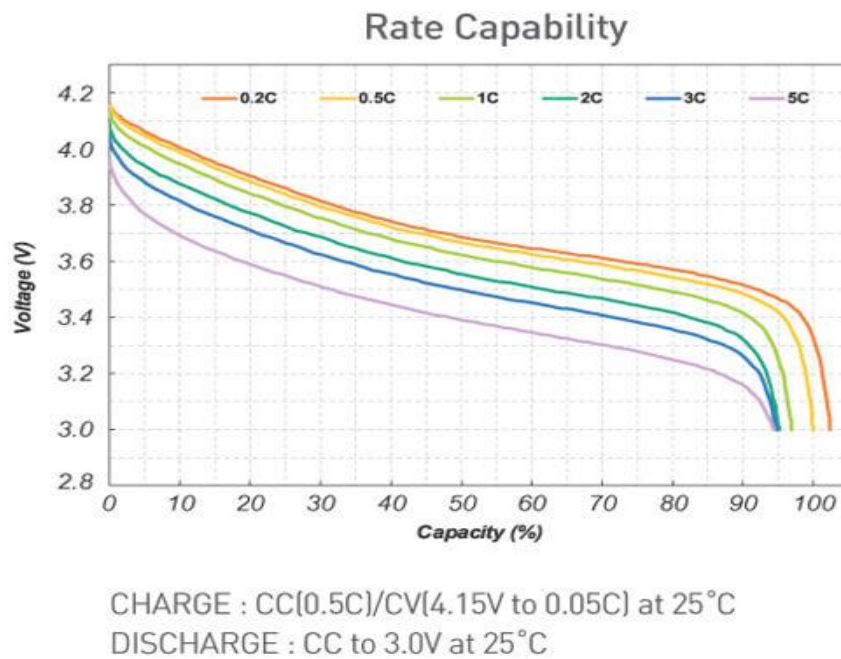


Figure.2.13 High rate characteristics [25]

The correct charging of a battery based on lithium polymers is essential for the durability and to avoid any type of accident or failure that could put at risk the safety of the person who handles it. The voltage of a cell of this type of battery, as mentioned above, is 3.7 Volts, this voltage is not constant, and it varies according to the percentage of charge when the battery is at 100% is 4.2 Volts. So, in order to charge the LiPo batteries we need to select the right battery with the right charger, majority of chargers on the market have built-in a system that auto-detects the number of cells connected to avoid risks. To summarize, the correct charging of a LiPo battery is

essential for its durability and safety. The battery voltage should not exceed the overall voltage of the battery during charging, and a suitable charger must be used to select the correct number of cells [24].

2.16 Criteria for battery pack selection:

1. Number of cells: the number of cells is determined by the system requirement however the battery power should be compatible with the controller input voltage.
2. Choice of Capacity: The capacity of the battery pack should be chosen based on the desired flight time and the weight of the overall system. Increasing capacity can extend flight time, but it's important to know that larger battery capacity have extra weight that will affect overall performance.
3. Choice of Discharge Capacity: The discharge capacity of the battery pack should be sufficient to handle the maximum current consumption of the system. It's essential to ensure that the battery's continuous discharge rating is suitable for the maximum current drawn by the system. For example, if the system consumes 100A and the battery capacity is 5000mAh, a minimum continuous discharge capacity of 20C is required. ($10000\text{mA}/5000\text{mah} = 20$) However, selecting batteries with extremely high discharge capacities isn't always a plus because. Higher discharge capacities result in increased weight and resistance, leading to voltage drops. So, it's important to balance and choose a discharge capacity that meets the system's requirements without sacrifice. Selecting a discharge capacity of 25C, for example, for a system with a maximum consumption of 40A, can provide a margin of safety and avoid excessive weight and voltage drop.
4. Temperature Range: temperature range at which can operate safely and effectively, when recharging lithium-ion batteries can be charged at max rate within a small window 20-45 C, outside temp range volt/current need to be lowered, which will result in longer charging time. We all heard about lithium battery that explode and that may have happen because of overcharging, higher or lower temp charging or short circuit.

2.17 Global traction and control system:

After we talked about all the parts and flight requirements and taking into consideration the restrictions and part compatibility, we can connect all the parts together.

2.18 Connectors:

In order to make connections easier and standardized there are a lot of connections that can be used [17].

JST Connector shown in figure 2.14: A small connector used for requirements up to 5 amps nominal. Used in batteries of reduced capacity, below 1500 mAh, to power the motors of very small UAVs and the onboard electronics of the models of a higher range.

Dean Ultra connector: A widely used type of connector whose price is relatively high compared to its competitors, can withstand a rated current level of up to 50 Amperes.

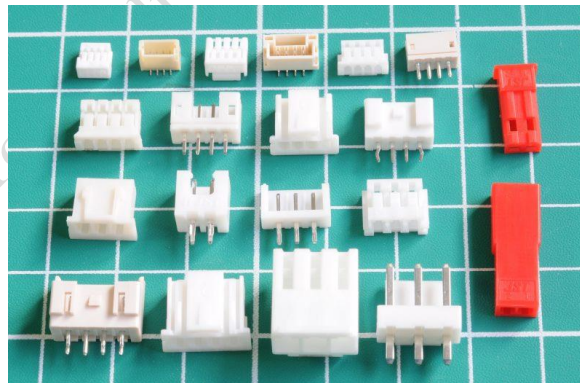


Figure.2.14 JST connector

EC3 connector: Bullet type connectors capable of withstand a continuous current of 60 Amps. The "bullet" type, seen in the picture, is very popular in high-power applications thanks to the fact that they have a larger contact surface.

EC5 Connector: Larger version of the EC3 shown in figure 2.15, with 5-millimeter pins, increases the contact surface thereby increasing its rated current capacity as it can reach up to 120 amperes.



Figure.2.15 EC3

Connector XT-60 shown in figure 2.16: One of the most popular connectors in the market for model aircraft thanks to its good value for money. Just as the EC3 connectors use "bullet" technology when making the connection and are capable of supporting currents of 65 Amperes. The protective case is made of nylon that does not melt when welding, and its shape makes them easy to connect and disconnect. There is a higher version able to withstand an intensity of up to 90 Amps at rated speed.



Figure.2.16 TX60

JST-XH connector: The most used connector for the cables to perform the rolling of the drone. This is shown in figure 2.17.



Figure.2.17 Jst-xh

One way to achieve compatibility is to do a research and see people experience in battery and connector and base your purchase on that.

2.19 Power Distribution Board:

The power distribution board is a critical component in multi-rotor UAVs, distributing power from the power system to the motors. A sample distribution board is shown in figure 2.18. It ensures that each motor receives the necessary power for optimal performance and stability, enabling flight maneuvers and control, some boards like the f3 EVO flight controller provide this service [27].

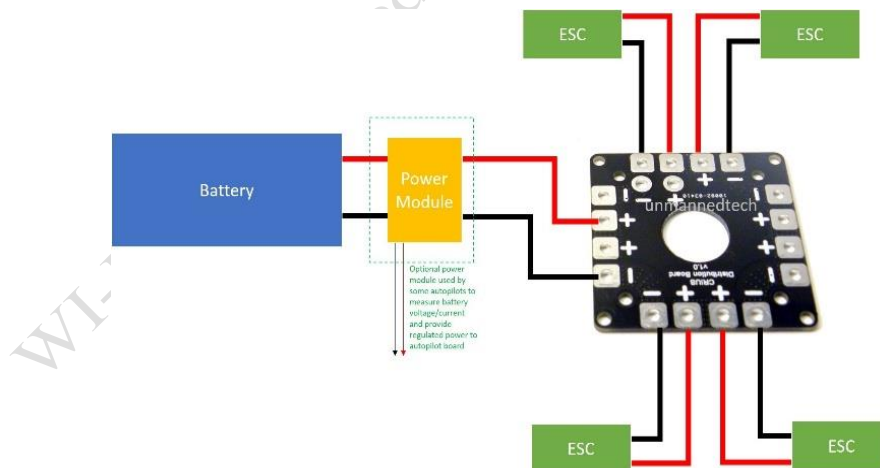


Figure.2.18 Power Distribution Board [27].

2.20 Flight Controller:

The flight controller is brain of the aircraft, responsible for controlling the engines to achieve desired movements. One of the flight controller samples is shown in figure 2.19. It consists of a circuit with inputs, outputs, and built-in sensors that determine the real-time position of the aircraft. The controller processes sensor data and address information using algorithms to send appropriate commands to the motors, typically using PWM signals.

The number and quality of sensors incorporated into the flight controller vary by model. Commonly used sensors include gyroscopes for orientation, Inertial Measurement Units for measuring speed, orientation, and other gravitational forces using accelerometers, gyroscopes, and magnetometers. Other sensors like sonar or lasers can provide distance information, and GPS units offer real-time geographical coordinates, enabling waypoint flights and automated routes.

The flight controller's capabilities depend on its design and complexity. More advanced controllers can perform complex flight maneuvers. The cost of the controller generally increases with its capabilities.

When choosing a flight control system, it's important to consider the ability to modify its programming. While most systems are closed and provide services as specified by the manufacturer, open-source devices and microcontroller boards offer flexibility for modifying behavior and adding new features.

In conclusion, the flight controller serves as the brain of the aircraft, utilizing sensors and algorithms to control the motors and perform various tasks, with options for customization and flexibility depending on the chosen system [17].

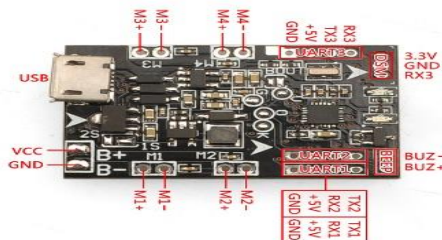


Figure.2.19 Flight Controller

Chapter 3

WIFI-controlled drone - Hardware design

3.1 introduction

In this chapter, we are going to explore the hardware design of Wi-Fi-controlled drones. We are going to cover essential components like the flight controller, controlling system, power delivery, and drone components. By understanding this, we can build reliable, efficient, and high-performing drones. The figure in 3.1 shows a depicted diagram of for WIFI controlled drone. The flight controller is going to be configured using Betaflight software and Blynk platform is going to be used to set up the buttons and slide buttons for the speed of the brushed rotors to control the drone direction or movement based on the required speed that initiate torques around the drone and consequently control movement. Also, configuration of WIFI through Arduino C software with required speeds of each motor. Home router or WIFI of cellphone can be used to connect the drone with the Blynk platform via internet. It is worth mentioning that the processing speed depends on the series of controller used.(i.e. F3, F4, or F7). Also, ESP8266 is connected with the flight controller receiver to receive commands and executed via flight controller to control the rotors speed and consequently the drone movement. Cellphone or IPAD is going to be used to send the commands via Blynk platform.

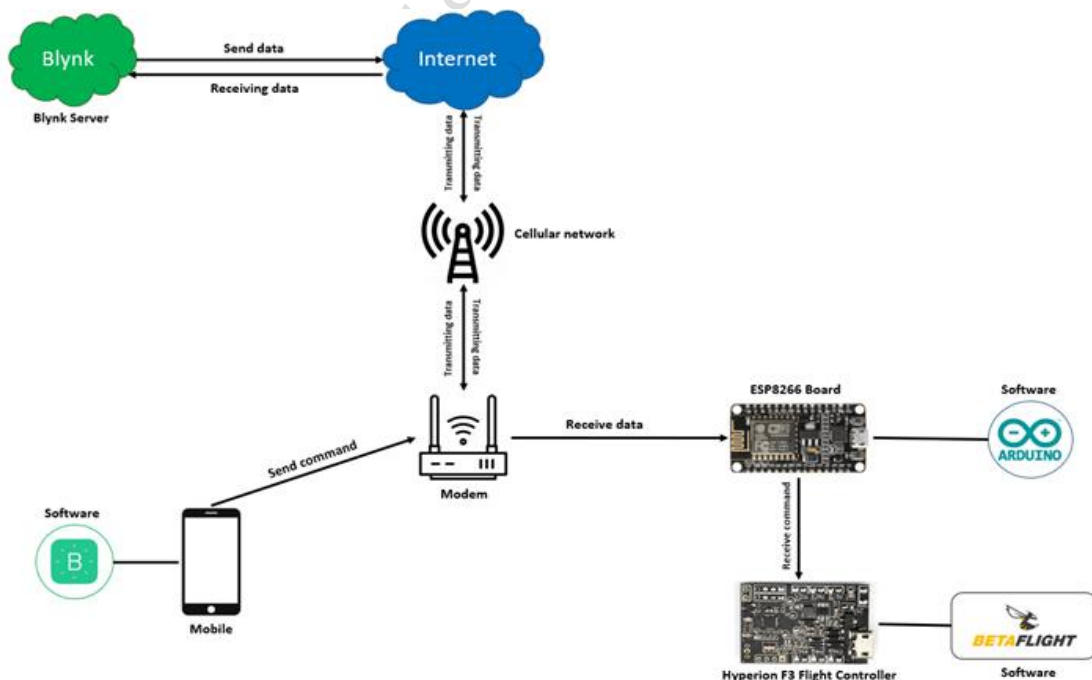


Figure 3.1 Connection chart

3.2 Design of Wi-Fi-controlled drone:

It is crucial to have a solid understanding of the many factors that affect a quad drone's operation and performance when designing one. Here are some crucial things to focus on [28].

3.2.1 Drone Frame Design

The drone's frame should be strong, lightweight, aerodynamically tuned, and robust. It should give enough room for mounting parts and still be stable when in flight.

3.2.2 flight Controller

It's important to choose the right flying controller. It should have strong control capabilities and support Wi-Fi networking. Also, it's important that it supports multiple receiver types such as sBUS or iBUS, PPM or PWM. The flight controller provides consistent flight performance, processes sensor data, and carries out flight commands [29]. SMT32 company is producing flight controllers series start with F3 to F7. Each one has his own features and the available series that is compatible with the design is going to be chosen. There are extra features added in each series that follow F3 such as presence of receiver communicating with satellites and transmitter to transmit signals from sensors such as camera, elevation measures, or processor speed of commands.

3.2.3 Communication System

For smooth drone operation, a dependable communication system is required. This entails picking suitable Wi-Fi modules and antennae as well as making sure that signals are transmitted and received correctly.

3.2.4 Power Management

Flight requires effective power management. To guarantee that all components receive enough supply of power, it is crucial to select the appropriate batteries, voltage regulators, and power distribution systems.

3.2.5 Propulsion System

Choosing the appropriate motors, propellers, and ESCs (if needed) is essential to obtaining the required maneuverability and flight characteristics.

3.2.6 Circuit Diagram

In the circuit design shown in figure 3.2, we are going to use the ESP8266 Wi-Fi Module as a receiver to control the drone, also we are going to use a 4 dc motor that should provide the necessary thrust, propulsion, and maneuverability, and of course, a flight controller that is the brain of the drone and connect all drone parts together, and a LiPo battery to provide the necessary power.

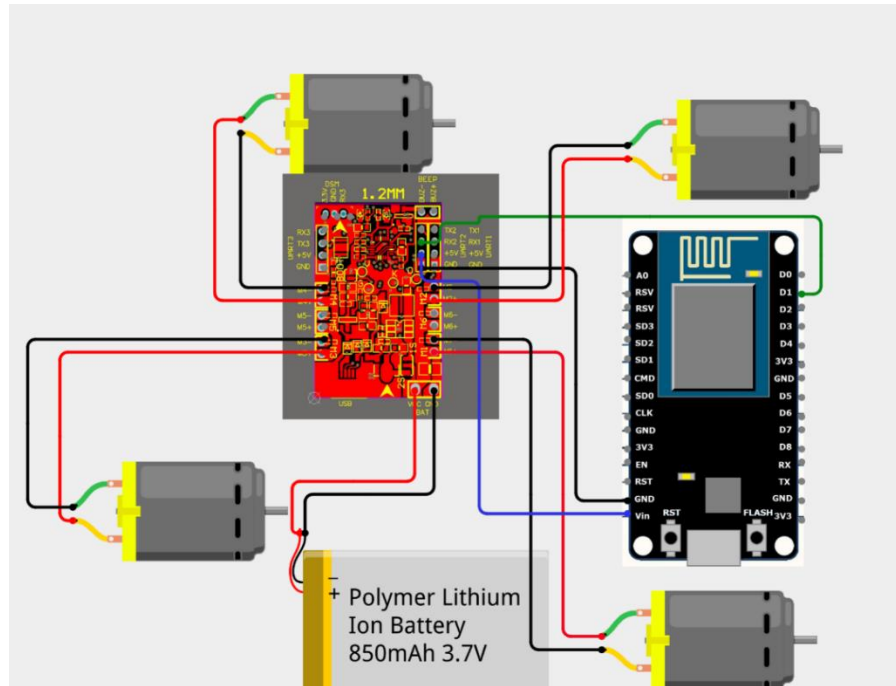


Figure 3.2 Circuit diagram [30].

Table 3.1: PIN connection table

component	Component pins	Flight controller pins
LiPo Battery 1S	+, -	VCC, GND
DC motor	P2 - P1+ P1+ P2-	M2, M3 M1, M4
ESP8266	Vin, GND, D1	5V, GND, RX2

3.3 Parts used in the Wi-Fi controlled drone

In the following, we are going to explore the parts of the drone.

3.3.1 F3 EVO Brushed V2.0 Flight Controller

Features:

1. The Hyperion F3 EVO Brushed Flight Controller, a 32-bit brush flight controller built on SP RACING F3 EVO firmware, is included in the revision based on the F3 EVO SP RACING.
2. This flight control system is designed to work with both 1S (4.2V) and 2S (8.4V) power sources, providing compatibility for different power supply configurations.
3. It incorporates an independent circuit structure design with pressure reduction technology. It can handle input from either a 1S or 2S power source and provides 3.3V or 5V outputs for UART1/2 and UART3.
4. Improved stability in flight is provided because of the modern hardware architecture F3, which uses STM32F303CCT6 + MPU6500.
5. NMOS transistors with six-way huge current capability can operate at up to 10 amps or more. Each device contains diodes that allow for freewheeling.
6. Support for receiver input signals from PPM, SBUS, and DSM.
7. Featuring a buzzer interface and battery voltage detection.
8. Six motors are supported.

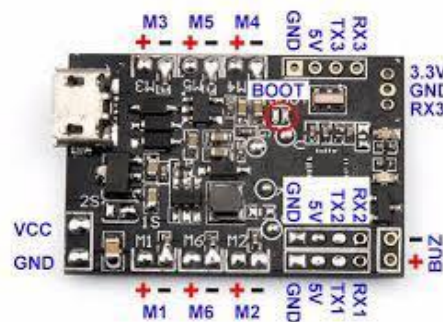


Figure.3.3 Pins of Hyperion F3.

Pins:

Micro USB socket: is employed to link a computer and flight controller, this enables you to update the flight controller's firmware.

M1-M6: Connect to Brush motor.

VCC: The flight controller is connected to the battery or power source via the VCC connector.

GND: The flight controller is connected to ground via the GND connector.

UART1: Numerous devices may be connected to the F3 EVO Brushed V2.0 Flight Controller via UART1.

UART2: The drone can be controlled by connecting your receiver to UART2.

UART3: usually employed when attaching an OSD.

BUZ+ -: The flight controller and buzzer's negative lead are connected via this pad.
 DSM/DSM2/DSMX: Protocols for Spektrum Receivers.

3.3.2 ESP8266 NodeMCU CP2102 Board

The CP2102 chip allows easy programming and debugging through a USB connection. The board has GPIO pins for connecting external devices and sensors, and it can be programmed using the Arduino IDE or Lua scripting language. It supports popular Wi-Fi standards and protocols for IoT communication.

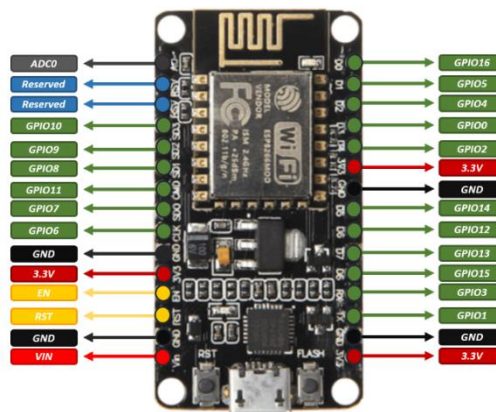


Figure.3.4 ESP8266 board pin layout.

Pins:

Micro USB socket: connect the USB port to power the NodeMCU.

GND: connect the NodeMCU board to ground.

3.3V: Anything that is connected to it receives 3.3V power from it.

EN, RST: resets microcontroller

ADC [analog pin]: utilized to measure analog voltage between 0 and 3.3V

RSV: Reserved

GPIO0-GPIO16: can be set up to function as PWM outputs, inputs, or outputs.

Vin: External Power Source

3.3.3 Coreless 8520 motors

This motor is DC brushed motor. It is a simple and cost-effective electric motor that uses brushes and a commutator to operate. It's easy to control but requires maintenance due to brush wear. It provides the necessary thrust, propulsion, and maneuverability.

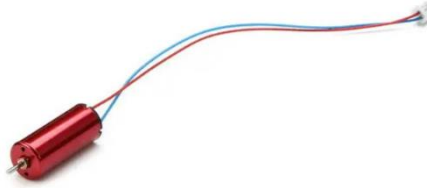


Figure.3.5 Coreless 8520 motor

3.3.4 Q100 Mini

Drone frames are the structures that hold drone components together and made of lightweight materials and provide stability and attachment points for motors, propellers, and other parts.



Figure.3.6 Q100 Mini

3.3.5 Propeller

Propellers are a vital component of a drone's propulsion system, working with the motors to generate thrust for flight. They come in various types, differing in sizes, materials, and their rotation direction and pitch affect performance. Proper balancing and installation are crucial for smooth operation. Safety considerations, such as using propeller guards, should be taken into account. Understanding propellers' role and characteristics enables drone builders to optimize thrust, efficiency, and overall flight performance.



Figure.3.7 Propellers

3.3.6 CNHL 2S Lipo Battery 7.4V 1100mAh

The flight controller of a drone typically uses a 1S or 2S battery to deliver power to the entire system. A 1S battery (3.7V) is common for smaller drones, however, if you want higher power and you don't have a problem in spending more money some flight controllers support a 2S battery (7.4V) which provides more power for larger drones. Factors such as battery capacity(mah), discharge rate(cc), and weight should be considered. Proper power management, including voltage regulation and fail-safe mechanisms, is crucial for optimal performance and battery lifespan.



Figure.3.8 CNHL 2S Lipo Battery

Chapter 4

WIFI-controlled drone - Software Overview

4.1 INTRODUCTION:

For this project, we are going to use a variety of software to control the drone and enhance its functionality for example configuring motors, receiver ports, and power also in our case using Wi-Fi to control the drone we are going to use the Arduino IDE to program the Wi-Fi board, also we are going to use an IoT platform to control the drone Via mobile device platform such as blynk.

4.2 Flight controller software

An essential part of drone control systems is flight controller software. It is in charge of overseeing and managing all aspects of the drone's flight operations, including navigation, energy monitoring, stabilization, and receiving user input, for example, programs clean flight and Betaflight all provide these services[31].



Figure.4.1 Flight controller software (firmware)[31].

4.2.1 Betaflight

This is defined as a flight control software, used for flying multi rotor radio controlled aircraft. Originating from the Baseflight and Cleanflight open-source projects. It has all

the functionality needed to control drones and configure it also it has a good user interface[32].

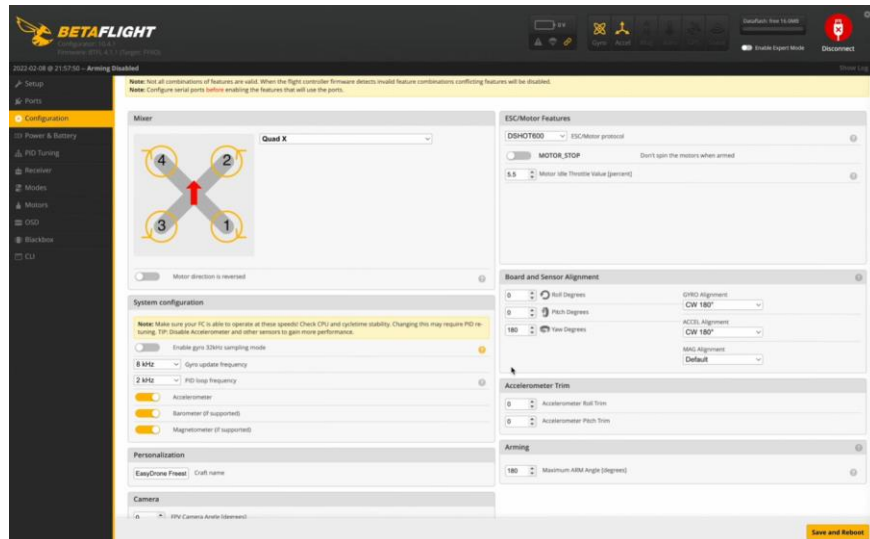


Figure.4.2 Betaflight software[32].

4.3 IoT platform

IoT platforms connect devices, manage data, and enable application development for better, connected solutions in various fields, and we are going to use them to control the drone[3].



Figure.4.3 IoT cloud platforms[33].

4.3.1 Blynk

Blynk is an open-source platform. It provides a user-friendly mobile app and cloud infrastructure to connect, monitor, and control IoT devices, and it will be used in this as a transmitter to control the drone[34].



Figure.4.3 Blynk Software[34].

4.4 Arduino IDE

Arduino Integrated Development Environment (IDE) is an open-source Arduino software. It provides a simple interface for writing, compiling, and uploading code to Arduino boards. It uses C++ programming language and has libraries, it will be used in this project to program the ESP8266 board and connect it to Wi-Fi[35].



Figure.4.4 Arduino IDE Software[35].

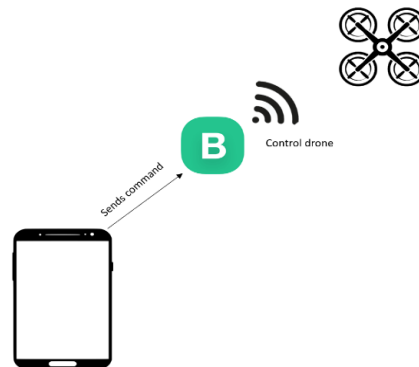


Figure.4.5 Connection diagram

4.5 Flowchart

In this flowchart when a user wants to fly the drone they will open the Blynk app which connects via internet to use the ESP8266 as a virtual controller. The ESP8266 receives commands from the user and transmits them to the flight controller, the flight controller will receive the data via ppm. Then check if the drone motor is not armed they will wait for the user command to arm it, but if the drone is armed the flight controller fires the motors and starts flying according to the user's command and configuration to the flight controller.

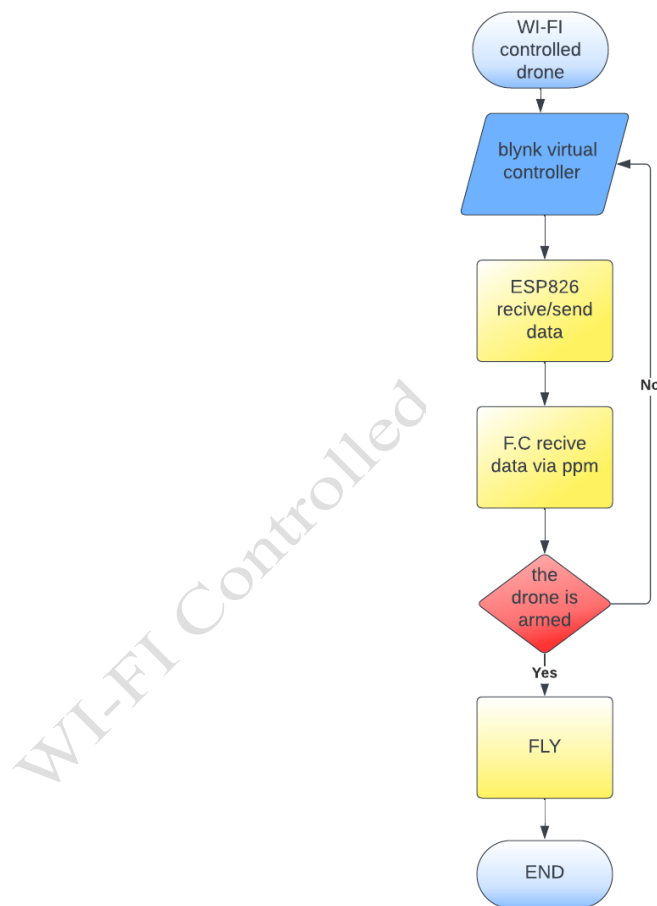


Figure.4.6 Project flowchart

Chapter 5

Design modificationsan Hardware construction

Due to many reasons, the design has been modified to suit the nature of the project and obsolescence of some materials. Part of these reasons are the following:

1. Lack of the control card for Hyperon F3 Acro in the market and searching for a replacement from different dealers.
2. The flight controller F4 Hyperon Acro has extra capabilities with a little bit higher price than the flight controller F3.
3. The length of time for obtaining materials through the carrier from outside Saudi Arabia.
4. The necessity to find an interface to connect with wireless network 4G/5G using internet protocols.
5. ESP8266 was used as a communicator between wireless 4G/5 G and the flight controller but it has some problems that hinder the operation of the motors specially the compatibility of the Baud rate between the ESP8266 and the flight controller card.
6. The ESP8266 was replaced with ESP32 to suit baud rates and overcome the problems with ESP8266.
7. The necessity to produce pulse position modulation (PPM) to direct control F4 flight controller via ESP32 and produce different pulses for Yaw, Pitch, Roll, and thrust.
8. The original design in part of the project, brushed motors were used and replaced with brushless motors as it is easier to operate and have control on the speed compared to brushed motors.

The single line diagram of the project is shown in figure 5.1. It consists of flight controller F4 Hyperion, ESP32, four ESCs, 4 brushless motors, Batteries, four propellers, and drone frame.

The motors have been chosen to lift the weight of the drone. In the following sections, we are going explain the details about the design components.

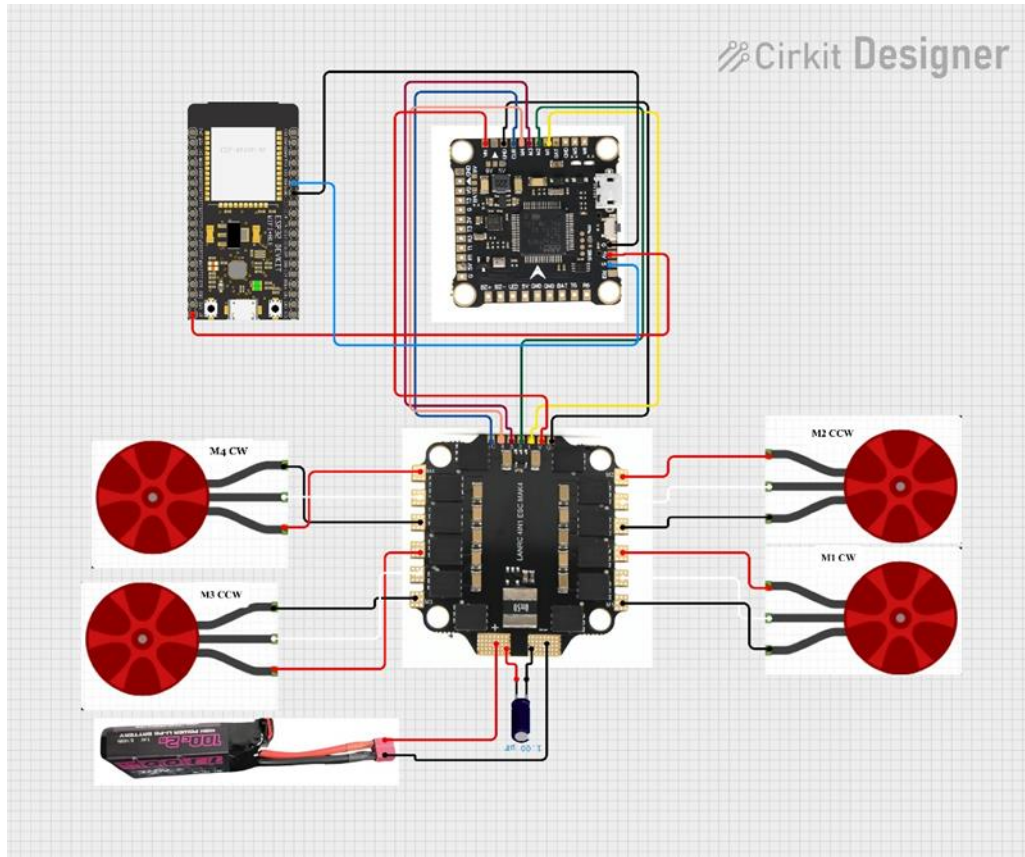


Figure 5.1: The single line diagram of Wi-Fi-controlled drone

Table 5.1: PIN connection table

Component	Component pins	PDB/4in1 ESC	Flight Controller
LiPo Battery 2S	+, -	+, -	
Brushless motor	+, S, -	M1, M2, M3, M4	
ESP32	GIOP21, 5V, GND		S(signal), 5V, G(Ground)
Flight Controller	CUR, M4, M3, M2, M1, V, G	C, 4, 3, 2, 1, V, G	

5.1 Flight Controller (LANNRC Omnibus F4 V3S PLUS Flight Control)

The flight controller is the central nervous system of the drone. It processes input from the receiver and sensors, making real-time adjustments to the motor speeds via the ESCs, ensuring stability and responsiveness. The LANNRC Omnibus F4 V3S PLUS is equipped with advanced sensors and firmware that support complex flight algorithms, which are crucial for maintaining flight control and handling various flight modes.

we connect the fc to the receiver(ESP32) via the pins S(signal),5V, G(Ground) which connected to ESP32 pins GIOP21,5V, GND respectively, and we need shorten the 2 PPM pins in order for the PPM signal to work which is highlighted Figure 5.3, it also connected to the PDB/4in1 ESC via 8pin JST SH 8pins (Japan Solderless Terminal)C,4,3,2,1, V, G Connected to CUR, M4, M3, M2, M1, V, G. The fc is shown in Figure 5.2.

Specification:

Microcontroller: MCU F40X

CLK speed: 160MHZ

GYRO=MPU6500

Have 3 UARTs port

Support PPM,sBUS,iBus protocols [36]

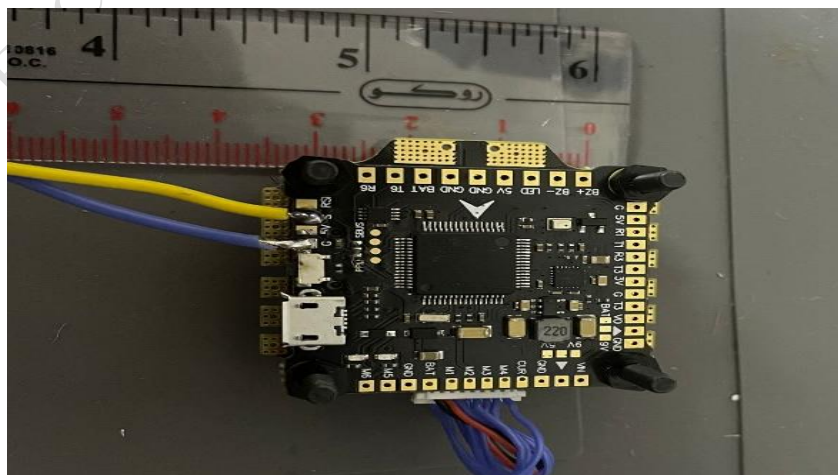


Figure 5.2: LANNRC Omnibus F4 V3S PLUS Flight Control)

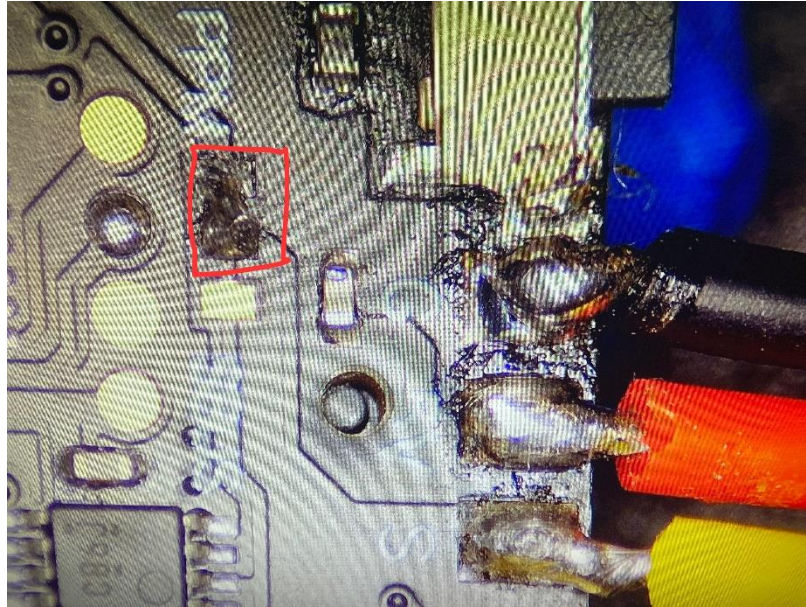


Figure 5.3: shows the 2 shorted PPM PINS

5.2 Power distribution board (PDB)

The Electronic Speed Controller (ESC) receives throttle signals from the flight controller and regulates the power sent to each motor. This 4-in-1 unit not only simplifies wiring by combining four ESCs into one board but also integrates a Power Distribution Board (PDB), which distributes power from the battery to the flight controller, ESCs, and other onboard electronics. This setup ensures efficient power management and minimizes weight, which is vital for flight performance. This is shown in Figure 5.4.

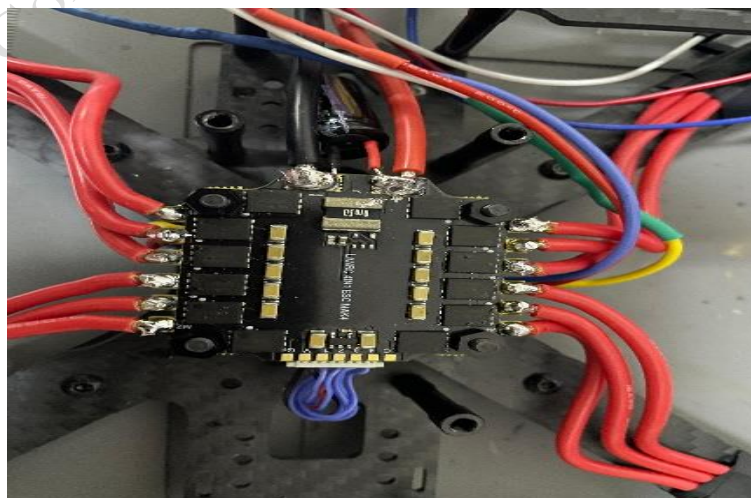


Figure 5.4: The ESC/power distribution board

5.3 Electronic speed controller (45A ESC)

The speed controller is an important part of the drone that is connected with the flight controller. Its main function is to transfer DC power to AC power that generates electromagnetic fields alternatively to maintain the stators of the motors revolving around the pivot which is consequently rotates the fans. It has two input wires and three output wires. These are power, ground, and PPM signals. Figure 5.4 shows the ESCs and the connections to motors and batteries.

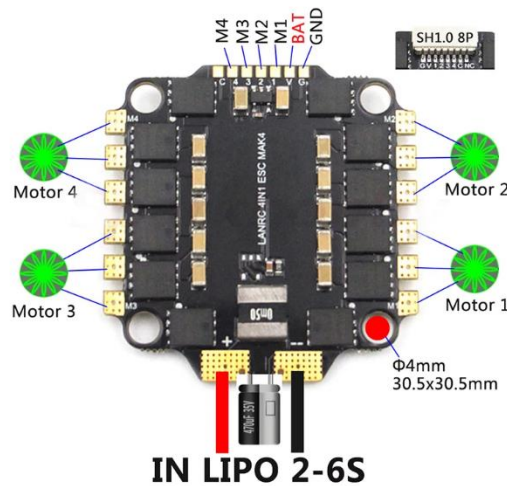


Figure 5.5: The 45A 4IN1 ESC

5.4 Receiver (ESP-WROOM-32 ESP32 ESP-32S)

The receiver acts as the communication bridge between the drone and the remote controller. The ESP-WROOM-32 is known for its robust Wi-Fi capabilities, allowing for additional functionalities AND providing good conductivity and extra pins that can be utilized in future projects. In our project, it works as an RC receiver that receives commands from the pilot via Blynk app and passes them to the flight controller using PPM signal generation, one of its functions is to generate a PPM signal, which then executes these commands, its connected to the FC using 3 pins GIOP21,GND,Vin 5v

connected respectively to FC, S(signal), GND,5V, uses bud rate of 115200 also the fc uses the same bud rate. The board is shown in Figure 5.5[37].

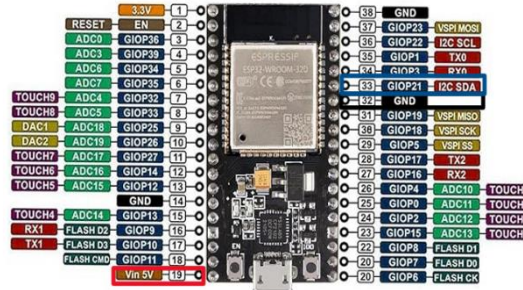


Figure 5.5: ESP32

5.5 Battery (1100mAh 7.4V 2S 100C)

The battery provides the necessary power to all electronic components of the drone, including motors and onboard electronics. battery capacity is 1100mAh which means it can deliver 1.1 Amp for an hour, depending on the voltage of the battery that's 8.14Wh, it also have A high discharge rate (100C) means the battery can be discharged at 100 times its capacity mean $(C \cdot A) = \text{Max Current}$ $(100 \cdot 1.1) = 110A$, allowing the battery to release energy quickly when needed, providing the motors with sufficient power to perform rapid maneuvers and maintain stability in the air. The battery is connected to the PDB/4in1 ESC where it will deliver necessary power the motor. The battery is shown in Figure 5.6.



Figure 5.6: Batteries used for the drone.

5.6 Brushless Outrunner Motor

There are two types of motors. These are brushed and brushless motors. In this project, brushless motors are used with ESCs to have more control on speed and give high thrust to drone.

5.6.1 Brushed vs Brushless Motors

When choosing between brushed and brushless DC motors for drones, there are several factors to consider that generally make brushless motors the better choice:

Efficiency and Power: Brushless motors are more efficient than brushed motors. They convert electricity into power more effectively, which is crucial for the power-to-weight ratio in drones.

Durability: Brushless motors have a longer lifespan. They don't have brushes that wear out, which reduces maintenance needs and improves reliability over time.

Performance: Brushless motors provide higher speed and torque. They are also capable of more precise control, which is essential for the stability and maneuverability of drones.

Heat Generation: Brushless motors generate less heat compared to brushed motors. Excess heat can be a significant issue in drones, affecting performance and potentially damaging components.

Cost: The primary advantage of brushed motors is their lower cost. However, for drones, where efficiency, performance, and durability are more critical, the higher initial cost of brushless motors is usually justified.

5.6.2 Brushless motor A2212 2450KV 5T

Brushless motors are preferred in drones for their efficiency and longevity and packing of high-power. They usually have high KV(constant voltage), KV indicates high RPMs at a given voltage, for example motor with 2450KV will have 2400RPM per given volts. ideal for driving the propellers at high speeds. motors convert electrical energy from the battery into mechanical energy, which spins the propellers to create thrust. the motor is shown in Figure 5.7.



Figure 5.7: Motors used with the drone

5.6.3 how to choose the correct motor

For choosing a motor you need to have the volume of a cylinder $= \pi r^2 h$, and you get that from writing on the motor AXXYY. The first 2 digits stand for the Cylinder diameter $r = \text{Cylinder diameter}/2$, the last 2 digits stand for the height of the cylinder h , XT T stands for turns around the digit before it stands for how many times the internal wires are wrapped around the armature lower turns have higher RPM and less torque, the KV stands for constant voltage the number before it means rpm per volt in fpv drone we use values from 1400KV to 2700KV depending on the drone weight, props, the thrust needed and battery voltage you can also decrease it from the PID suction in beta flight by decreasing motor output, if you have an x inch quadcopter x stands for propeller size:

Propeller size 5 cylinder volume $\geq 1884 \text{ mm}^3$

Propeller size 6 cylinder volume $\geq 2189 \text{ mm}^3$

Propeller size 7 cylinder volume $\geq 2826 \text{ mm}^3$ [38]

5.7 Frame (3K Full Carbon Fiber XL7)

The frame is the structural backbone of the drone, providing mounting points for all components and protecting them during flight. Carbon fiber is chosen for its strength-to-weight ratio, essential for durability and agility. The XL7 design likely offers good aerodynamics, which helps in improving the overall flight performance. The frame is shown in figure 5.8.



Figure 5.8: The frame used for the drone

5.8 Propeller (HQPROP 7X3.5X3 7035 3-Blade Poly Carbonate CW CCW Propeller)

Propellers are the thrust-generating element of the drone. The "7035" designation refers to the size (7 inches in diameter with a 3.5 pitch), optimized for balance between thrust, efficiency, and control. These propel the drone through the air and are directly influenced by the motor's speed, which is controlled by the ESC based on commands from the flight controller. It is shown in Figure 5.9.



Figure 5.9: Propellers used in the project[39]

5.9 Integration and Functionality

In operation, the pilot's commands are relayed by the ESP32 receiver to the flight controller, which interprets these commands to control the drone's behavior through precise motor speed adjustments made by the ESCs. The integration of these components allows for complex control mechanisms, enhancing the drone's performance in flight. The structural arrangement in the diagram ensures efficient wiring and component placement, optimizing space and minimizing the potential for electrical failures, crucial for maintaining the drone's performance and reliability during flight.

Chapter 6

Software Design

Introduction to the Software Used in the Drone Project

a combination of sophisticated software platforms and tools has been used to manage and control its operation effectively. In the following sections, the software used to control the drone is explained.

6.1 Software diagram

For The software diagram first the Data from the user's mobile phone Blynk app to the user modem then it will go through the internet to Blynk server which will send the data to the Esp32 through the internet the esp32 works as a receiver will then take the data and run the code that generates PPM signals and will send it to the Flight Controller witch will use the Betalight firmware to process the data and output the proper value to the motors. The Blynk allows using control buttons and these buttons send the signal to ESP32 via used identified GSM network with internet protocols to control the speed of propellers using the produced 4 types of PPM signals from ESP32 to the flight controller. These PPM 4 signals are controlling the thrust, YAW, PITCH, and roll of the drone. This is show in figure 6.1.

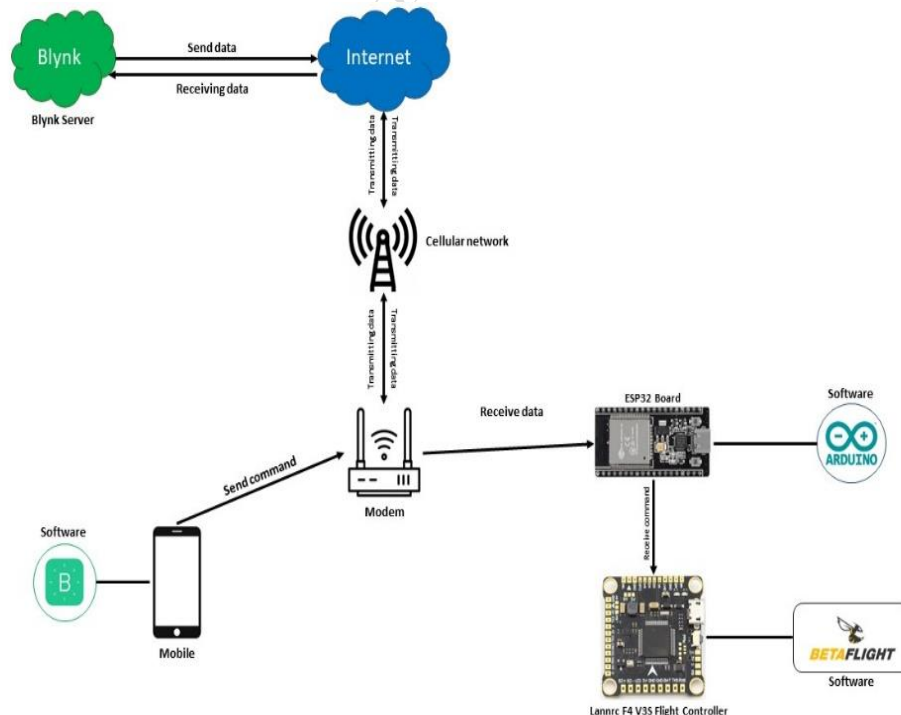


Figure.6.1: WIFI controlled drone software diagram

6.2 Flowchart

In this flowchart when a user wants to fly the drone they will open the Blynk app which connects via the internet BLYNK virtual controller and the ESP32 as a receiver commands sent from the user will be sent to ESP32 which will turn it into a proper ppm signal the Flight controller will check the gyroscope and the Receiver values, then it will process it After that will check if the arm channel is on or off if it's off the fc will loop back to the previous step, otherwise if its armed the fc will calculate PID(Proportional (P), Integral (I), Derivative (D)) the FC will output the values to each motor via the ESC, then it will loop back to check the arming step. This is shown in 6.2.

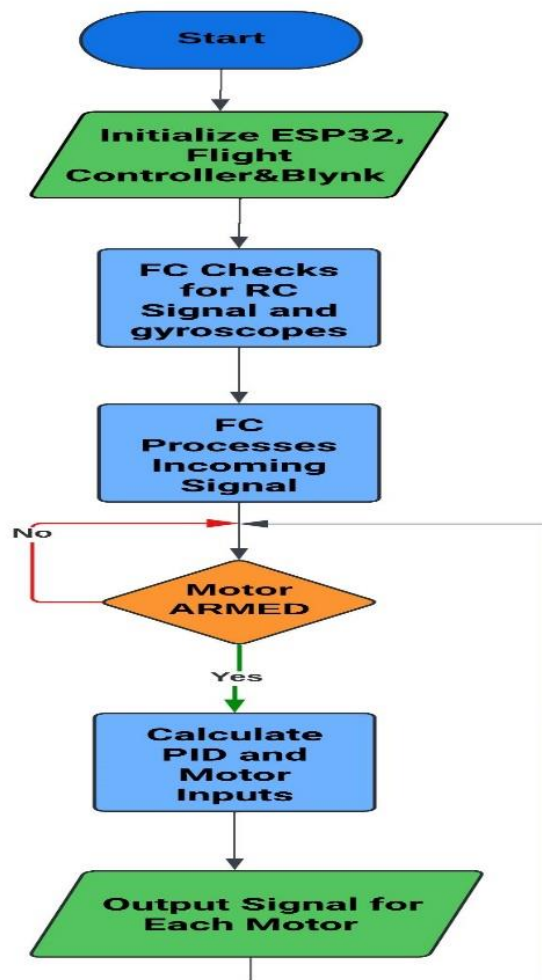


Figure.6.2: Project Flowchart

The network used is defined in Arduino environment commands with the PPM out signals using wireless internet protocols. The program is given in Appendix A.

6.3 Betaflight Firmware

Betaflight is an open-source firmware designed for flight controller boards used in drones and other similar devices. It offers highly configurable settings and is optimized for performance with low-latency communication. It's ideal for a variety of multirotor configurations and is widely recognized for its stability and comprehensive support for different types of sensors and motors. The front window of the Betaflight is shown in figure 6.3.

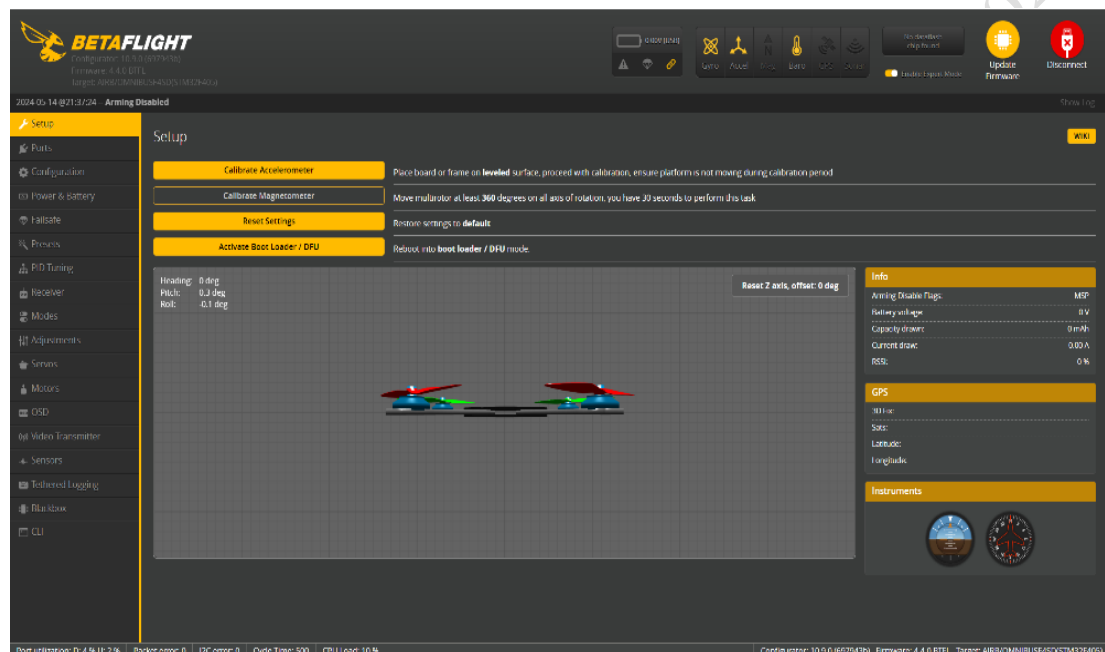


Figure.6.3: Betaflight software

6.3.1 setting up betaflight

Betaflight provides a lot functions however most of it depend on the flight controller to support it I will provide a small guide to setup Betaflight. First of all you need to download the betaflight app to your computer after doing that you need to download the necessary the drivers that you will find the link for in the opening page for Betaflight.

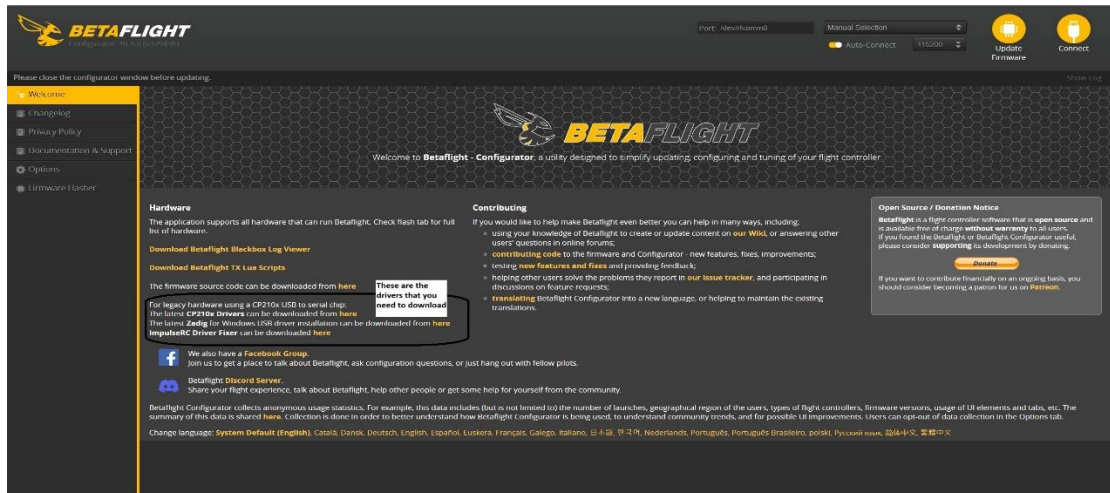


Figure.6.4: Betaflight front panel

6.3.2 Betaflight Taps

After downloading the app and the necessary drivers you need to flash the fc and download recent firmware via Firmware Flasher tab.

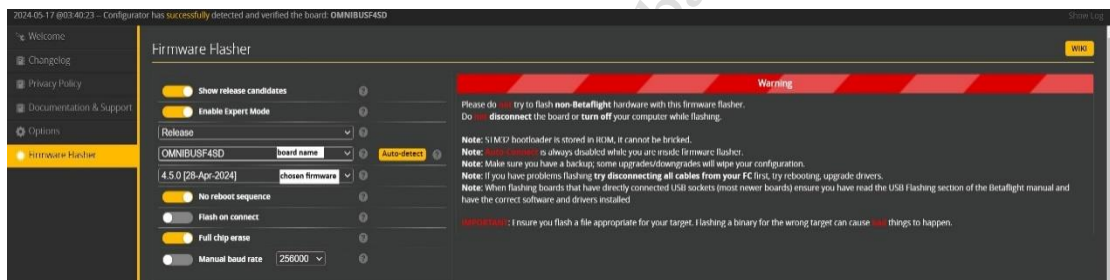


Figure.6.5: Betaflight Firmware flasher tab

After that we can start working on the fc we need to setup the Ports tab in order to communicate with the ESP32 by setting UART3 to serial RX and the USB VCP is set automatically to talk with the computer.

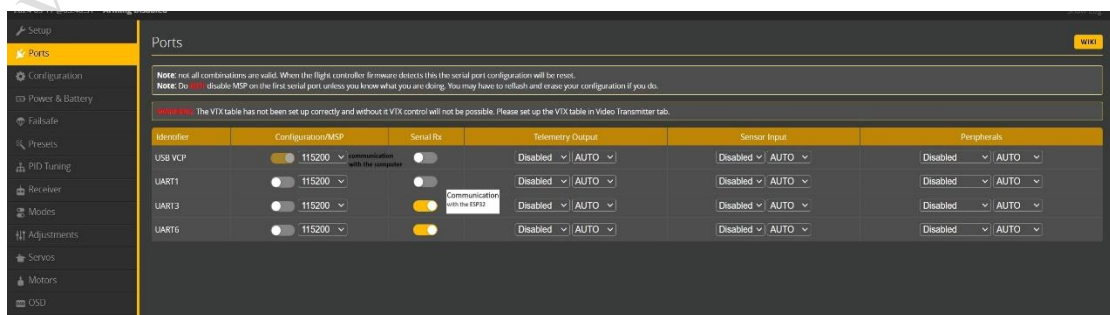


Figure.6.6: Betaflight Ports tab

The configuration tab and power battery tabs are usually set automatically by the FC after that we go to the Fail Safe tab and setup the quad to land when fail-safe is triggered.

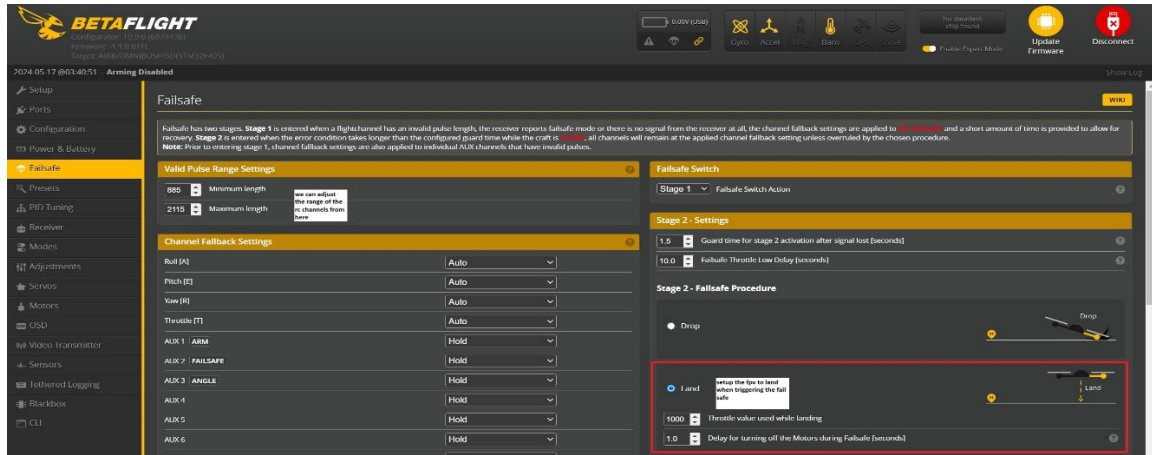


Figure.6.7: Betaflight Fail safe tab

After that, we go to the PID tuning section in this tab we can change the motor output power and the PID parameter which is usually advised to not change unless there is a problem.

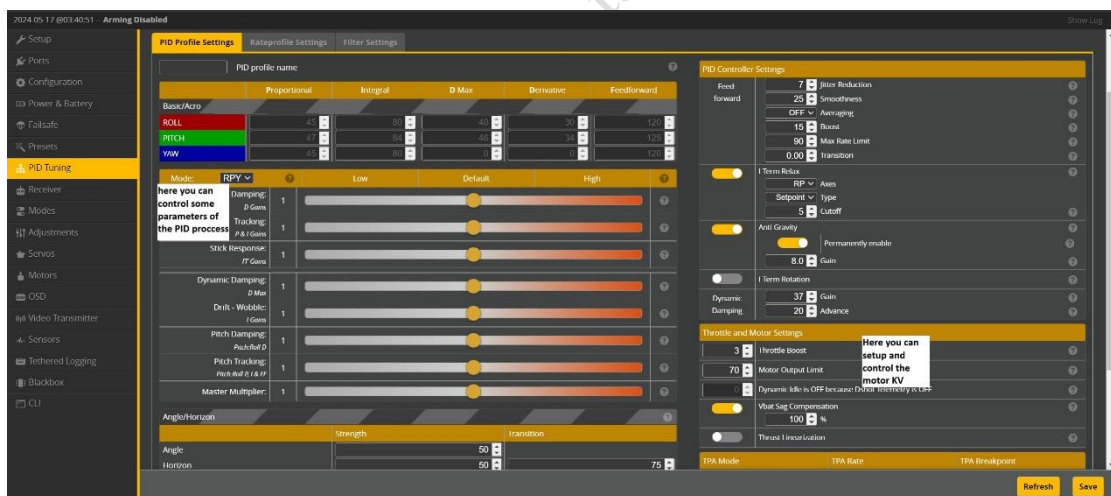
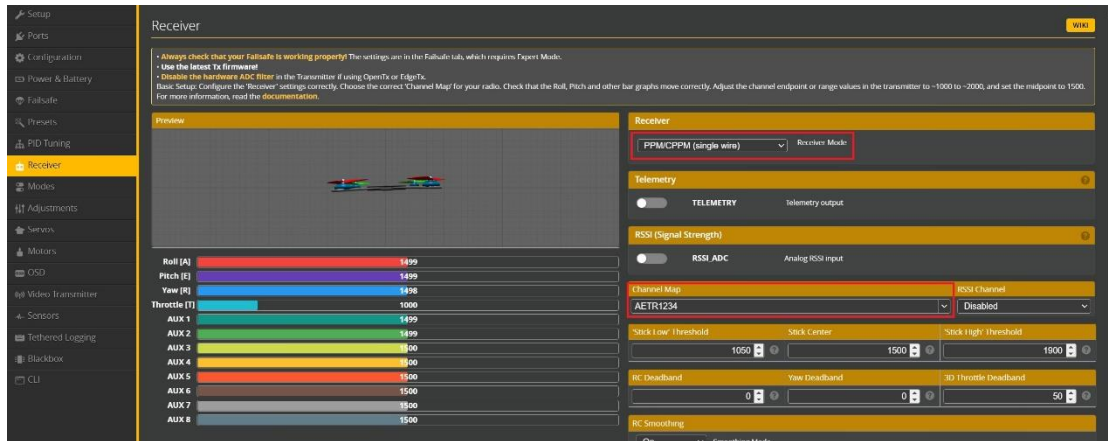


Figure.6.8: Betaflight PID tuning tab

After that from the receiver tab we make sure we choose the receiver mode to be PPM and if we connect the Esp32 we can check if the RC is working by watching the values. Also we need to make sure that Channel Map is AETR1234 as it is the ESP32 in order for the value to be mapped correctly.



Then we go to Modes tab to setup the mode we need in our Case ARM to arm the quade. Angle mode which should make the drone to self level when you move your hand from the fc although recently it had some issues in the betaflyght firmware, and the Fail safe channel to be able to land the drone when needed or a problem accurse.



Figure.6.10: Betaflight Modes tab

now the betaflight should be set.[40]

6.4 Arduino IDE

The Arduino Integrated Development Environment (IDE) is used for writing and uploading code to the flight controller hardware. It is a popular choice in the hobbyist and maker communities due to its user-friendly interface and extensive library support. In this project, Arduino IDE allows you to program the ESP-WROOM-32 module, enabling it to function as the receiver and communicator with the Blynk app. needs few

things to communicate with Blynk platform TEMPLATE_ID, TEMPLATE_NAME, AUTH_TOKEN, Wi-Fi and password of the used network and the necessary libraries. The program used is shown in Appendix A. [35].



Figure.6.11: Arduino IDE Software[35].

In this project, we are using the Arduino IDE to write code to the ESP32 but after downloading the program, it is needed to be set up. Then, downloading the specific library for a specific board and in this project, ESP32 is used.

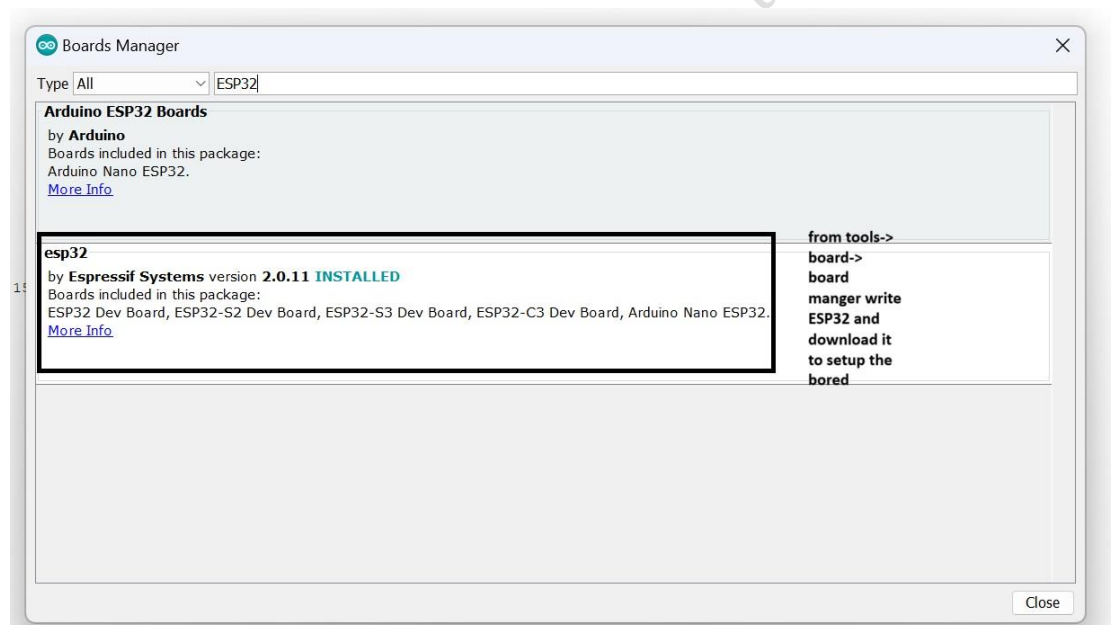


Figure 6.12: Downloading ESP32 library

Part of the set up for the Arduino ESP32 library is choosing the COM port and the operating frequency. Figure 6.13 shows the window of setting up ESP32 on Arduino platform. Also, it is important to download the BLYNK library to be able to connect the BLYNK with ESP32 which is shown in figure 6.14. This can be done via using Tools library, manager type (Blynk).

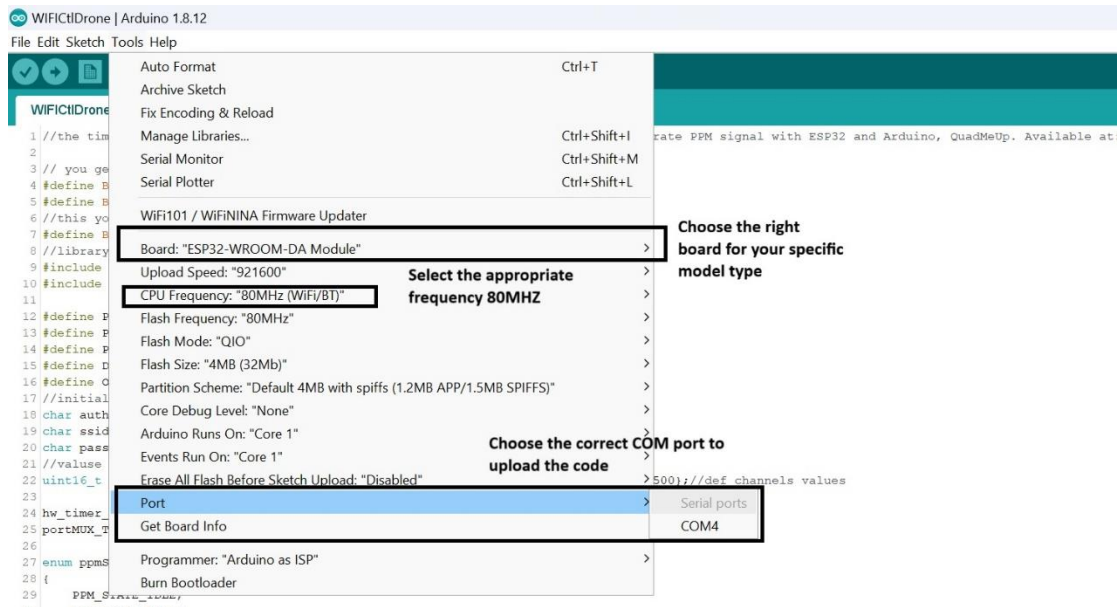


Figure 6.13: Window showing the set up of ESP32 in Arduino platform

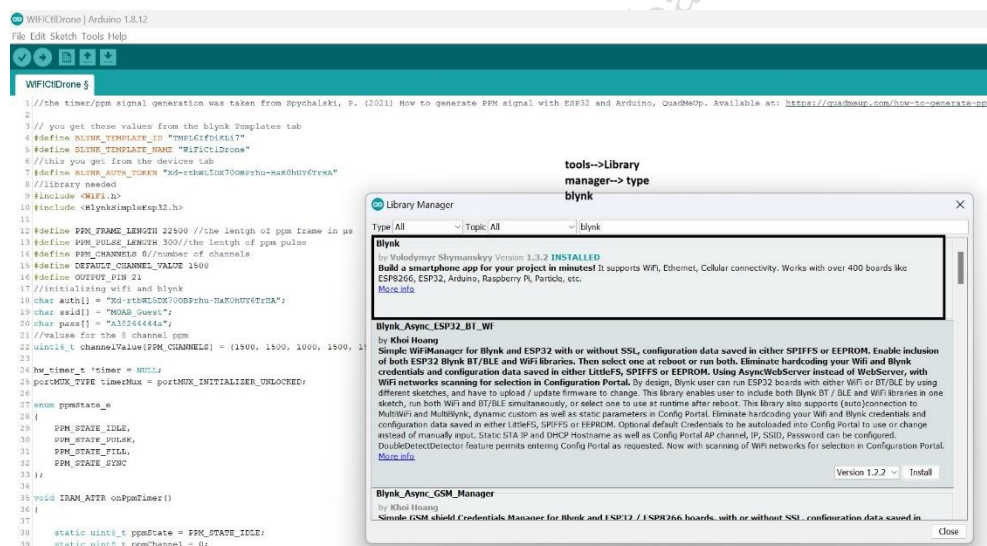


Figure 6.14: Window showing installation of Blynk manger library

6.5 Blynk IoT Platform

Blynk is a digital platform that offers a user-friendly way to build mobile applications for controlling IoT devices. In the drone project, Blynk provides the interface to control the drone remotely, sending commands over the Internet or local network which are then processed by the ESP32 receiver. [34].



Figure.6.15: Blynk Software

To control the drone via the internet we need to use an IOT platform such as blynk. Starting BLYNK platform starts by making an account and logging then creating a template Tab as can be seen in figure 6.16.

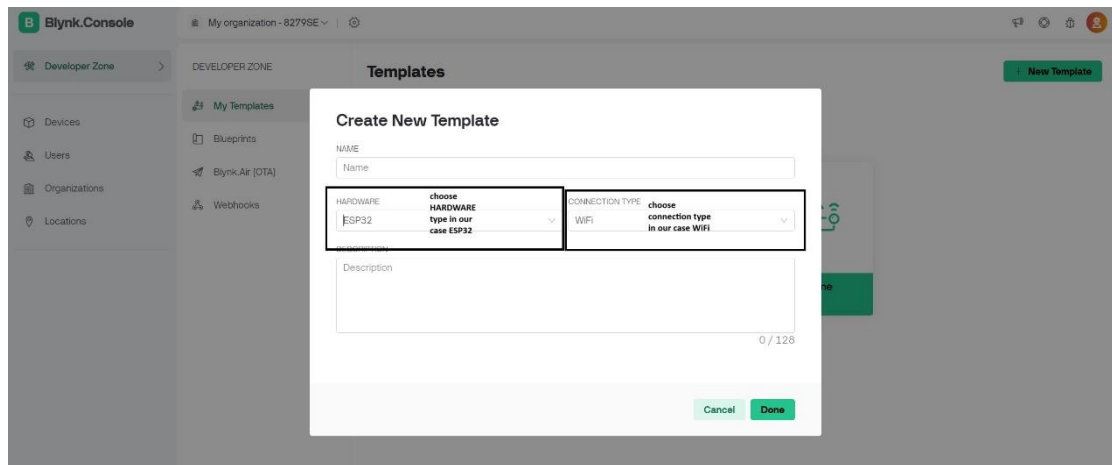


Figure 6.16: starting BLYNK platform by making an account and logging in

After this step, you are eligible to get template ID (BLYNK_TEMPLATE_ID) and template name (BLYNK_TEMPLATE_NAME). Then, you are authorized to create a device as can be seen in figure 6.17. At this stage, there are plenty of devices and in our case, the drone device should be chosen.

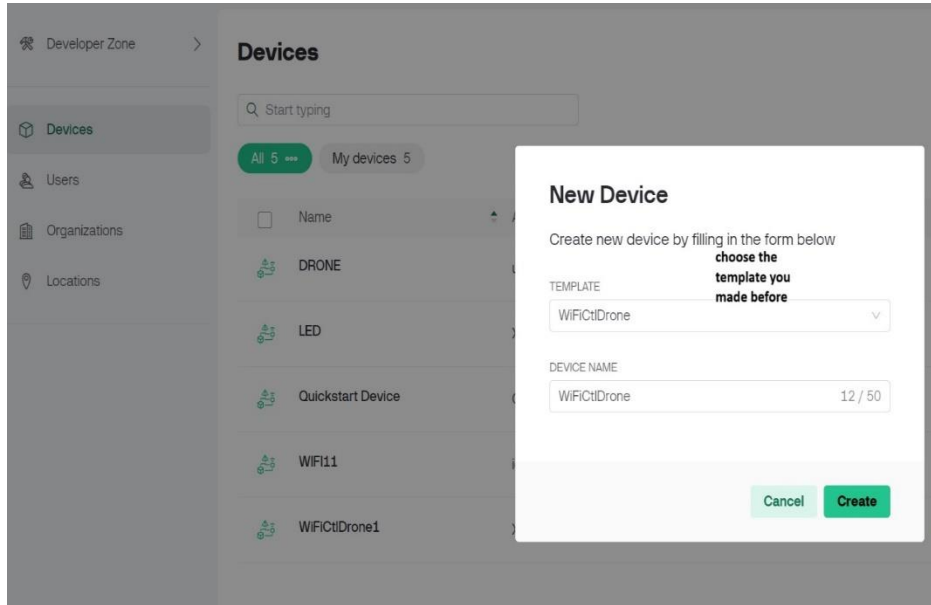


Figure 6.17: Authorization to make a device and drone is chosen.

After making the device , and to create all virtual pins needed, click on the data stream then new data stream. Name each virtual pin and choose the virtual path then save it. This should be repeated for each virtual pin needed for controlling the drone via the mobile application, namely, throttle, pitch, Yaw, and Roll. This can be seen in figure 6.18.

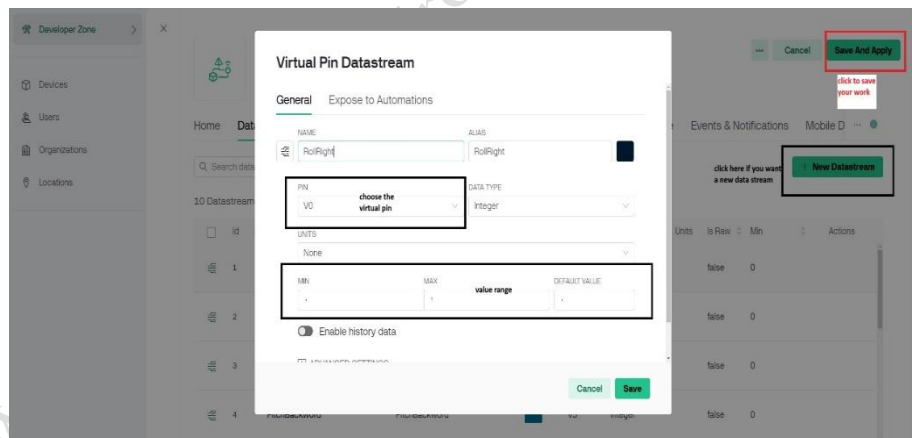


Figure 6.18: Naming and setting up the virtual pins needed for the device (drone)

After creating all virtual interface pins needed to control the drone, it is required to log in into the inbox and choose the device you created as can be seen in figure 6.19 and click on the plus button to choose the widgets you want to appear for controlling as can be seen in figure 6.19 and 6.20 respectively.

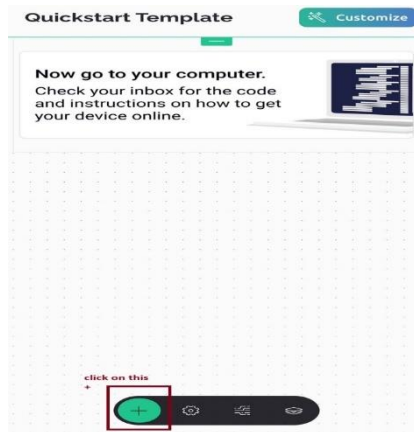


Figure 6.19: The quick start template after logging in into inbox

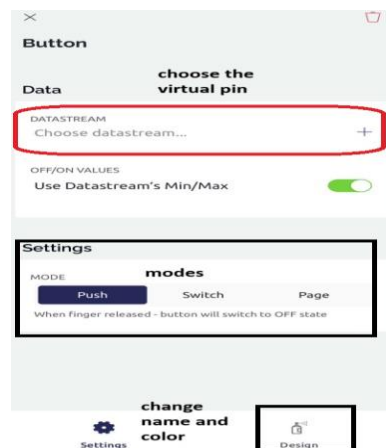


Figure 6.20: Setting up the widgets

Also, figure 6.21 shows an Example for an interface.

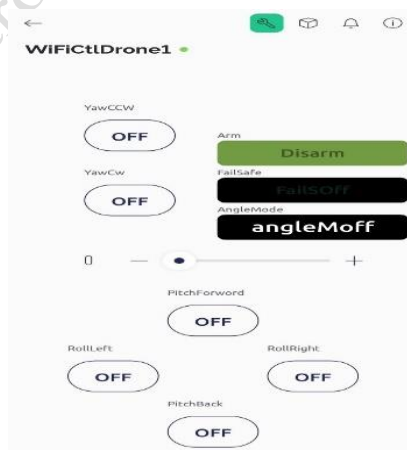


Figure 6.21: Example of an interface

6.5.1 Control Interface Setup

User Interaction: The Blynk application serves as the primary interface for the pilot. The app sends control commands (such as altitude, speed, and direction) through the network.

6.5.2 Data Transmission

Network Communication: Commands from the Blynk app are transmitted over the Internet or local Wi-Fi to the ESP-WROOM-32 receiver. The ESP32, programmed using Arduino IDE, is set up to receive these commands and forward them to the flight controller that has been set by connecting a virtual data stream for the required input. These are shown clearly in figure 6.22 and figure 6.23 respectively.

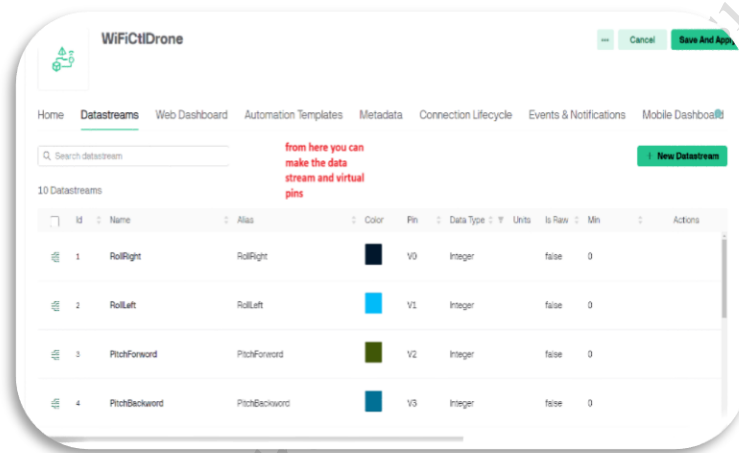


Figure.6.22: Blynk data stream

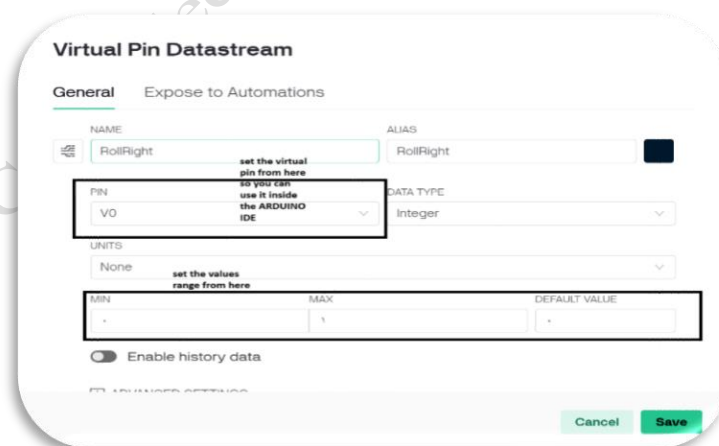


Figure.6.23: Blynk Virtual Pins

6.5.3 Command Processing

Flight Controller Action: The flight controller, running Betaflight firmware, interprets the commands received from the ESP32. It processes this data, factoring in readings

from onboard sensors (like gyroscopes and accelerometers) to maintain stability and control.

6.5.4 Motor Control

ESC and Motor Response: Based on the processed commands and sensor data, the flight controller sends signals to the ESC, adjusting the speed of each motor. This precise control over motor speed is what allows the drone to change altitude, direction, and speed.

6.5.5 Propeller Adjustment

Physical Movement: Changes in motor speed result in adjustments to the propeller speeds, causing the drone to move according to the pilot's commands sent from the Blynk app.

6.6 BLHeli firmware

It is a popular firmware that is used to control ESCs and configure them for example starting power and temp protection , you can also choose the motor spin direction without changing the wire orientation[41] as can be shown in figure 6.24.

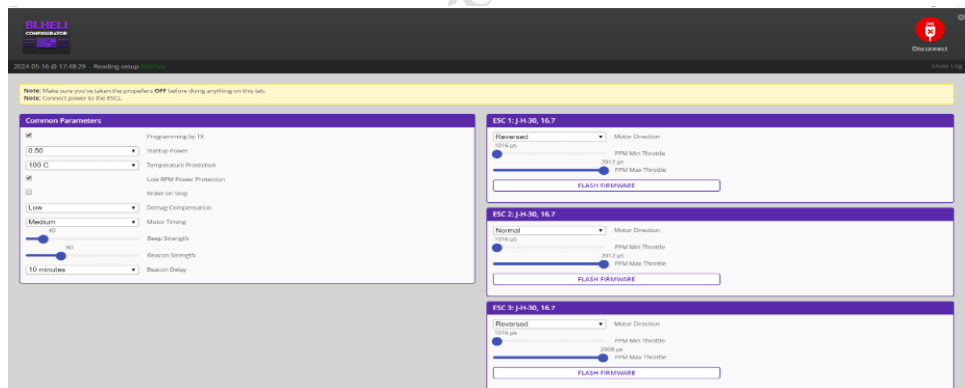


Figure.6.24: BLHeli interface

Chapter 7

Conclusions and Future work

7.1 Conclusions

This project is about controlling a drone using WIFI connection. The first part in designing the project, a brushed motors were intended to be used. However, using brushless motors were found to give higher speed and more control for the drone. The availability of materials in the market such as the flight controller F3 was found to be scarce because the factory has developed the controller and added extra features to newer versions of the flight controller such as F4 till F7. Also, the other factor was the transportation of materials from outside Saudi Arabia that took sometimes to start working on the project. Moreover, the minimum baud rate for the flight controller is 115200 bps and the baud rate of ESP32 should be compatible with the flight controller. The drone used has high specifications that can be used in racing speeds.

7.2 Future work

The first series of these projects started with controlling the drone. However, this drone could have plenty of applications that can be used in future work. This drone can be used in civilian and military applications.

7.2.1 Civilian Applications

A camera can be mounted on the drone and videos can be transmitted to laptop or cellphone to do plenty of applications such as face recognition, image processing, traffic light violations, tracing traffic violators, vegetation monitoring, tracking lost people in desert, connecting drones to satellites to track stolen cars, they can be used in infected areas with viruses

to study the situation. Also, it can be used in Hajj and Omrah to directly monitoring the crowds, and many more applications.

7.2.2 Military Applications

This kind of high speed drones can be used in civil defense during floods, to monitor the traffic, issuing tickets for speed violations, and monitoring traffics. Also, it can be used in remote areas to monitor borders intruders, and helping soldiers during war to monitor the movements of enemies and neutralize the enemy danger.

WI-FI Controlled drone taibah university 2024

Appendix A

Arduino IDE

CODE:

//the timer/ppm signal generation was taken from Spsychalski, P. (2021) How to generate PPM signal with ESP32 and Arduino, QuadMeUp. Available at: <https://quadmeup.com/how-to-generate-ppm-signal-with-esp32-and-arduino/>. [42]

```
// you get these values from the blynk Templates tab
#define BLYNK_TEMPLATE_ID "TMPL6IfDiKLi7"
#define BLYNK_TEMPLATE_NAME "WiFiCtlDrone"
//this you get from the devices tab
#define BLYNK_AUTH_TOKEN "Xd-rtbWL5DX70OBPrhu-HaK0hUY6TrHA"
//library needed
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

#define PPM_FRAME_LENGTH 22500 //the lentgh of ppm frame in µs
#define PPM_PULSE_LENGTH 300 //the lentgh of ppm pulse
#define PPM_CHANNELS 8 //number of channels
#define DEFAULT_CHANNEL_VALUE 1500
#define OUTPUT_PIN 21
//initializing wifi and blynk
char auth[] = "Xd-rtbWL5DX70OBPrhu-HaK0hUY6TrHA";
char ssid[] = "Wi-Fi ssid";
char pass[] = "password";
//valuse for the 8 channel ppm
uint16_t channelValue[PPM_CHANNELS] = {1500, 1500, 1000, 1500, 1500, 1500, 1500, 1500}; //def channels values

hw_timer_t *timer = NULL;
portMUX_TYPE timerMux = portMUX_INITIALIZER_UNLOCKED;

enum ppmState_e
{
    PPM_STATE_IDLE,
    PPM_STATE_PULSE,
    PPM_STATE_FILL,
    PPM_STATE_SYNC
};

void IRAM_ATTR onPpmTimer()
{
    static uint8_t ppmState = PPM_STATE_IDLE;
    static uint8_t ppmChannel = 0;
    static uint8_t ppmOutput = LOW;
    static int usedFrameLength = 0;
    int currentChannelValue;

    portENTER_CRITICAL(&timerMux);

    if (ppmState == PPM_STATE_IDLE)
    {
        ppmState = PPM_STATE_PULSE;
```

```

        ppmChannel = 0;
        usedFrameLength = 0;
        ppmOutput = LOW;
    }

    if (ppmState == PPM_STATE_PULSE)
    {
        ppmOutput = HIGH;
        usedFrameLength += PPM_PULSE_LENGTH;
        ppmState = PPM_STATE_FILL;

        timerAlarmWrite(timer, PPM_PULSE_LENGTH, true);
    }
    else if (ppmState == PPM_STATE_FILL)
    {
        ppmOutput = LOW;
        currentChannelValue = channelValue[ppmChannel];

        ppmChannel++;
        ppmState = PPM_STATE_PULSE;

        if (ppmChannel >= PPM_CHANNELS)
        {
            ppmChannel = 0;
            timerAlarmWrite(timer, PPM_FRAME_LENGTH -
usedFrameLength, true);
            usedFrameLength = 0;
        }
        else
        {
            usedFrameLength += currentChannelValue -
PPM_PULSE_LENGTH;
            timerAlarmWrite(timer, currentChannelValue -
PPM_PULSE_LENGTH, true);
        }
    }
    portEXIT_CRITICAL(&timerMux);
    digitalWrite(OUTPUT_PIN, ppmOutput);
}

//setting up virtual pins VX v stands for virtual pin X stand for
the number of the v pin function to connect blynk interface to
arduino IDE
BLYNK_WRITE(V0) {
    int input = param.asInt(); // Get the current state of the button

    if (input == 1&&channelValue[0]<=1900) {
        channelValue[0] += 50; // increase 'Roll' by 50 when the
button is pressed. You can control movement speed by
increasing/decreasing this value from the code
        Serial.print("Roll: ");
        Serial.println(channelValue[0]); //serial print will be
printed to the serial monitor
    }else
        channelValue[0] = 1500;
}

```



```

        Serial.print("RollNoIF: ");
        Serial.println(channelValue[0]);
    }
    BLYNK_WRITE(V1) {
        int input = param.asInt(); // same as before

        if (input == 1 && channelValue[0] >= 1000) {
            channelValue[0] -= 50; // decreases Roll by 50 when the
            button is pressed
            Serial.print("Roll: ");
            Serial.println(channelValue[0]);

        } else
            channelValue[0] = 1500;
            Serial.print("RollNoIF: ");
            Serial.println(channelValue[0]);
        }
    BLYNK_WRITE(V2) {
        int input = param.asInt(); // same as roll function

        if (input == 1 && channelValue[1] <= 1900) {
            channelValue[1] += 50;
            Serial.print("Pitch: ");
            Serial.println(channelValue[1]);
        } else {
            channelValue[1] = 1500;
            Serial.print("PitchNoIF: ");
            Serial.println(channelValue[1]);
        }
    BLYNK_WRITE(V3) {
        int input = param.asInt();

        if (input == 1 && channelValue[1] >= 1000) {
            channelValue[1] -= 50;
            Serial.print("Pitch: ");
            Serial.println(channelValue[1]);
        } else {
            channelValue[1] = 1500;
            Serial.print("PitchNoIF: ");
            Serial.println(channelValue[1]);
        }
    BLYNK_WRITE(V4) {
        int input = param.asInt();
        channelValue[2] = map(input, 0, 250, 1000, 1900);
        Serial.print("Throttle: ");
        Serial.println(channelValue[2]);
    }
    // control yaw ccw and cw direction
    BLYNK_WRITE(V5) {
        int input = param.asInt();
        if (input == 1) {
            channelValue[3] = 1500 - 150;
            Serial.println(channelValue[3]);
        }
    }

```

```

else {
channelValue[3]=1500;
Serial.println(channelValue[3]);
}}
BLYNK_WRITE(V6) {
int input = param.asInt();
if(input==1){
channelValue[3]=1500+150;
Serial.println(channelValue[3]);}
else {
channelValue[3]=1500;
Serial.println(channelValue[3]);
}

}

// setting up the modes ARM,FailSafe,AngleMode

BLYNK_WRITE(V7) {
channelValue[4] =param.asInt();
Serial.print("arm: ");
Serial.println(channelValue[4]);

}
//set FailSafe mode
BLYNK_WRITE(V8) {
channelValue[5] =param.asInt();
Serial.print("Fail Safe: ");
Serial.println(channelValue[5]);
}
// set ANGLEMODE
BLYNK_WRITE(V9) {
channelValue[6] =param.asInt();
Serial.print("ANGLEMODE: ");
Serial.println(channelValue[6]);

}

void setup()
{
Serial.begin(115200);
Blynk.begin(auth, ssid, pass);

pinMode(OUTPUT_PIN, OUTPUT); //set the desired output pin
//setting up the timer
timer = timerBegin(0, 80, true);
timerAttachInterrupt(timer, &onPpmTimer, true);
timerAlarmWrite(timer, 12000, true);
timerAlarmEnable(timer);
}

void loop()
{
Blynk.run();

}

```

Safety measures

1. Battery safety

1.1 Charging and Discharging

Charging LiPo battery with a charger in particular designed for LiPo batteries. Avoid completely discharging LiPo batteries or discharging them to a point where each cell is under 3.5 voltage, as it can cause irreversible damage and reduce their lifespan. Use a LiPo voltage checker to monitor the battery voltage during use and land or recharge the battery before it reaches unsafe voltage levels. It is recommended to use a LiPo balance charger mode to charge each cell equally and have video display units.

1.2 Charging Area Safety

Charge LiPo batteries on a non-flammable floor, far from flammable materials, and don't leave them unattended while charging. LiPo batteries can generate heat all through the charging procedure, and in uncommon instances, this heat can lead to fire or explosion. It is really helpful to have a fire extinguisher with no trouble available as an important safety precaution.

1.3 Storage

Store LiPo batteries in a cool, dry place far away from flammable materials. It is beneficial to keep them in a fireproof LiPo protection bag or container designed for battery storage. LiPo batteries should be saved at around 50% charge capacity, as both high and low charge levels can degrade the battery's performance and safety.

1.4 Handling

Handle LiPo batteries with care and avoid physical damage, puncturing, or bending the battery. Always transport LiPo batteries in a safe and suitable container to prevent short circuits or accidental damage.

2. Propeller Safety

2.1 Avoid contact with propellers

Drones must never be operated indoors, near people, animals, or delicate objects to prevent potential injury or damage as a result of contact with spinning propellers.

2.2 Proper Installation

Ensure that propellers are securely attached to the drone's motors. Loose or improperly set up propellers can disconnect at some stage in flight, leading to risky flight behavior or maybe causing the drone to crash.

2.3 Inspection and Replacement

Regularly check out propellers for signs of damage, wear, or deformation. Cracks, chips, or great bending can compromise their structural integrity and lead to performance issues or failures. Replace any damaged or worn-out propellers with new ones of the ideal size and kind.

3. Motor Safety

3.1 Maintenance and Inspection

Regularly inspect at the drone's motors for any signs of damage, unfastened connections, or harm. Look for frayed wires, bent shafts, or excessive heat buildup. Address any issues promptly by repairing or replacing the motors as necessary.

3.2 Overheating Prevention

Motors can generate significant heat at some stage in operation. To save you from overheating, ensure that the drone isn't always operated continuously for extended intervals without allowing the motors to cool down.

3.3 Motor Failure Precautions

In the event of a motor failure all through the flight, the drone's stability and management can be compromised. It is crucial to practice emergency procedures, which include preserving calm, lowering the throttle steadily, and landing the drone safely. Develop situational attention and be organized to react appropriately if a motor malfunction takes place.

References

- [1] A brief history of drones Imperial War Museums. Available at: <https://www.iwm.org.uk/history/a-brief-history-of-dronesch>
- [2] Ruston, Proctor Radio Controlled Target Aircraft rae-o870.jpg. Available at: https://commons.wikimedia.org/wiki/File:Ruston,_Proctor_Radio_controlled_target_aircraft_RAE-O870.jpg.
- [3] Unmanned Aircraft Unmanned Aircraft - an overview | ScienceDirect Topics. Available at: <https://www.sciencedirect.com/topics/engineering/unmanned-aircraft>.
- [4] Single-rotor drones 360 Video Booth. Available at: <https://www.royalvideobooth.com/single-rotor-drones.html>.
- [5] Black Flying Multi Rotor Drone Sitting On The Ground Background, Small Drone For Aerial Photography
- [6] X8 flying wing - professional UAV by Airelectronics: Aeroexpo The B2B marketplace for aeronautical equipment. Available at: <https://www.aeroexpo.online/prod/airelectronics/product-185532-28782.html>.
- [7] CW-30E(CW-40) hybrid gasoline & Battery Long Flight Time UAV (2023) JOUAV. Available at: <https://www.jouav.com/products/cw-40.html>.
- [8] Advance, R. Home, High performance 4K DJI drone. Available at: <https://www.robot-advance.com/EN/art-drone-dji-mini-2-3632.html>.
- [9] Hossain, R. (2022) 'A Short Review of the Drone Technology ', International Journal of Mechatronics and Manufacturing Technology, pp. 60–66.
doi:https://www.researchgate.net/publication/362908663_A_Short_Review_of_the_Drone_Technology
- [10] Leslie, J. (2023) Benefits of using drones for construction inspections - 2021, Drones Survey Services. Available at: <https://dronesurveyservices.com/benefits-drones-construction-inspections/>.
- [11] Fitzsimmons, M. World of drones, New America. Available at: <https://www.newamerica.org/future-security/reports/world-drones/who-has-what-countries-with-armed-drones/>.
- [12] Leventhal, J. (2021) Storm drones could revolutionize weather forecasting, Nextgov.com. Available at: <https://www.nextgov.com/emerging-tech/2017/07/storm-drones-could-revolutionize-weather-forecasting/139251/>.
- [13] Butterworth-Hayes, P. (2019) Drone delivery operations underway in 27 countries, Unmanned airspace. Available at: <https://www.unmannedairspace.info/latest-news-and-information/drone-delivery-operations-underway-in-26-countries/>.
- [14] Drone shows in 2023 – Lunalite Drone Show Technologies (2023) LunaLite Drone Show Technologies – Drone Shows, Drone Light Show, Aerial Drone show performance. Available at: <https://lunalitedroneshow.com/drone-shows-in-2023/>.
- [15] The battle of quadcopters Quadcopter Arena. Available at: <https://quadcopterarena.com/>.
- [16] IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE) e-ISSN: 2278-1676,p-ISSN: 2320-3331, Volume 12, Issue 1 Ver. II (Jan. – Feb. 2017), PP 70-75 www.iosrjournals.org DOI: 10.9790/1676-1201027075 www.iosrjournals.org 70 | Page Theory and Analysis of Three Phase Induction Motor using Written Pole Technology
- [17] Herrero, R.A. 'DRONE DESIGN', DUMLUPINAR UNIVERSITESI INDUSTRIAL ENGINEERING, pp. 6–42. Available at:https://digibuo.uniovi.es/dspace/bitstream/handle/10651/43461/TFG_Antu%F1a_Herrero.pdf?sequence=3.

- [18] Different types of drone frames: Monocopter to Octocopter Different Types of Drone Frames in 2022: Monocopter to Octocopter. Available at: <https://circuitdigest.com/article/different-types-of-drone-frames-monocopter-to-octocopter>.
- [19] Hobbywireless. Available at: https://hobbywireless.com/racing-frames-only-c-178_266/nighthawk-x5-racing-frame-p-3953.html.
- [20] Lift IVAO Documentation Library. Available at: <https://wiki.ivao.aero/en/home/training/documentation/Lift>.
- [21] Rozum Brushless Motor: Rozum robotics, Rozum Robotics LLC. Available at: <https://rozum.com/brushless-motors/>.
- [22] Karnavas, Yannis & Chasiotis, Ioannis & Stravourellis, Dimitris. (2018). A Practical BLDC Motor Design Procedure for Diver Propulsion Vehicle Applications.
- [23] Estingoy, L. (2021) Inside the electronic speed control, Model Aviation. Available at: <https://www.modelaviation.com/inside-esc>.
- [24] A LIPO BATTERY GUIDE TO UNDERSTAND LIPO BATTERY GensTattu. Available at: <https://genstattu.com/bw/>.
- [25] High rate discharge Li Polymer Battery 1c~120c Li_Polymer_Battery_com. Available at: <https://li-polymer-battery.com/high-rate-discharge-li-polymer-battery-1c-120c/>.
- [26] Selecting the right battery for your application - important ... - ansys (2022) www.ansys.com. Available at: <https://www.ansys.com/blog/important-battery-metrics>.
- [27] unmannedtech1, A. (2015) Power Distribution Boards - How to choose the right one, DroneTrest. Available at: <https://www.dronetrest.com/t/power-distribution-boards-how-to-choose-the-right-one/1259>.
- [28] Step 1: Design your frame Step 1: Design your frame | Unmanned Aerial Vehicles Team. Available at: <https://sites.bu.edu/uav/first-build/step1/>.
- [29] What does IBUS, SBUS, PPM, PWM mean? (2021) Receiver Protocols Explained for FPV Drone Beginners (2021). RCwithAdam. 25 January. Available at: <https://youtu.be/y7T4hBqOwxM?si=UIOepiVkXHiJXV6G>.
- [30] gradProjectConnection (2023) cirkitsstudio.
- [31] Robert et al. (2023) Firmware for FPV drone flight controller overview, Oscar Liang. Available at: <https://oscarliang.com/fc-firmware/>.
- [32] Betaflight - pushing the limits of UAV performance: Betaflight Betaflight News RSS. Available at: <https://betaflight.com/>.
- [33] IOT cloud platforms IoT Design Pro - IoT Projects, Articles & News. Available at: <https://iotdesignpro.com/articles/top-10-iot-cloud-platforms>.
- [34] A low-code IOT software platform for businesses and developers Blynk. Available at: <https://blynk.io/>.
- [35] Team, T.A. *Software, Arduino*. Available at: <https://www.arduino.cc/en/software>.
- [36] ArduPilot, devanesanandrews Support for omnibus F4 v3s plus flight control · issue #21684 · Ardupilot/Ardupilot, GitHub. Available at: <https://github.com/ArduPilot/ardupilot/issues/21684>.
- [37] Anobium Anobium/Hiletgo: Hiletgo demonstrations, GitHub. Available at: <https://github.com/Anobium/HiLetgo>.
- [38] Spychalski, P. (2019) How to choose drone motors? kv and torque, YouTube. Available at: https://youtu.be/HAj_WLJkZg?si=jY--wKQ5PADheZEc.
- [39] https://www.aliexpress.com/item/1005006860201955.html?spm=a2g0o.order_list.order_list_main.53.42c21802jbaNqe.
- [40] Bardwell, J. (2022) Betaflight 4.3 install, flash, setup tab: Complete walkthrough, YouTube. Available at: https://youtu.be/LkBWREiEGKT?si=A4_aSPHbtwS_aJpI.
- [41] Bitdump Bitdump/blheli: Blheli for brushless ESC Firmware, GitHub. Available at: <https://github.com/bitdump/BLHeli>.
- [42] Spychalski, P. (2021) How to generate PPM signal with ESP32 and Arduino, QuadMeUp. Available at: <https://quadmeup.com/how-to-generate-ppm-signal-with-esp32-and-arduino/>.

Appendix-B

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Timeline (Gantt Chart)

